

Software Requirement Specification

COS 301 - Capstone 2026
Capstone Project: EquityLens
By The Big Five

Demo 4 version - 28 September 2026



Project Name: EquityLens
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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for EquityLens, an investment analytics platform. It is intended for the development team, project examiners and the stakeholders at AWS and it defines the scope and success criteria.

1.2 Business Need

Retail investing in South Africa has surged due to accessible brokerage platforms, but a significant analytical gap remains: retail investors have access to the market yet lack the tools needed to interpret risk metrics and connect world events to portfolio volatility. Investors face data overwhelm - they hold diversified portfolios but cannot easily see how concentrated they really are, or understand why specific stocks moved, without manual and error-prone spreadsheets.

EquityLens addresses this by centralising a user's investment data in a single dashboard. It imports brokerage statements, scores how well the portfolio is spread, calculates financial indicators, detects statistically unusual price moves in the user's holdings and links them to news published around each move, and uses a conversational AI assistant to explain the results in plain language which is bridging the gap between raw data and actionable insight.

1.3 Scope

EquityLens is a web-based analytics suite for South African retail investors. It covers:

- Secure registration with email verification, time-based one-time password (TOTP) multi-factor authentication, password reset and self-service account deletion
- Portfolio import from an EasyEquities PDF statement or the EquityLens Excel template, with each statement kept as a separate portfolio and a downloadable PDF snapshot report
- A dashboard that summarises portfolio value and today's move, a configurable structural health score, performance against a blended regional benchmark, positions and sector allocation, plain-language insights and sector what-if simulations
- Detection of statistically unusual daily price moves in the user's holdings, with a market-model explanation of each move and links to news published around it
- A news page for the user's holdings and for the general market
- Seven financial indicators per holding, each with a plain-language explanation
- A conversational AI assistant, available on its own page and from every signed-in page, with saved conversations, a user-managed memory and guardrails
- A watchlist for tracking shares, including ones the user does not own, and an Explore page that suggests JSE shares similar to those the user holds

Out of scope: direct stock trading, brokerage integration that requires the user's brokerage credentials, and regulated financial advice.

1.4 Definitions, Acronyms and Abbreviations

- EquityLens - the investment analytics web platform described in this document
- SRS - Software Requirements Specification

- SAS - Software Architecture Specification
- FR - Functional Requirement
- NFR - Non-Functional Requirement
- UC - Use Case
- US - User Story
- TUCBW - This Use Case Begins When
- TUCEW - This Use Case Ends When
- MFA - Multi-Factor Authentication
- TOTP - Time-based One-Time Password
- API - Application Programming Interface
- AWS - Amazon Web Services
- POPIA - Protection of Personal Information Act (South Africa)
- JSE - Johannesburg Stock Exchange
- JSE Top 40 - the 40 largest JSE-listed companies; the Satrix 40 exchange traded fund tracks this index
- ETF - Exchange Traded Fund
- LLM - Large Language Model
- CAPM - Capital Asset Pricing Model
- RSI - Relative Strength Index
- P/E - Price-to-Earnings ratio
- HHI - Herfindahl-Hirschman Index, the sum of squared weights, used to measure concentration
- EWMA - Exponentially Weighted Moving Average
- Unusual move - a daily price move at least three times the size of the holding's own recent typical daily move
- CAR - Cumulative Abnormal Return, the running sum of the part of a stock's return that the market does not explain
- TWR - Time-Weighted Return
- XIRR - money-weighted internal rate of return
- TFSA - Tax-Free Savings Account
- CGT - Capital Gains Tax
- ZAR - South African rand
- SAST - South African Standard Time (UTC+2)
- BM25 - a standard text-relevance ranking function used to order news articles
- CI/CD - Continuous Integration / Continuous Deployment

1.5 Document Overview

Section 2 describes the intended users and their goals as user stories. Section 3 presents the use case model: use case diagrams followed by a TUCBW/TUCEW description of each use case. Section 4 specifies the functional requirements, grouped into ten subsystems. Section 5 states the five targeted quality requirements, how each is verified and its current status. Section 6 presents the domain model as a UML class diagram with entity descriptions. Section 7 traces each use case to the functional requirements that realise it, and Section 8 records the assumptions and external dependencies the requirements rest on.

2. User Stories and User Characteristics

2.1 Intended Users

Retail Investor - the primary and only in-app user. A South African retail investor who imports portfolio data and uses the dashboard, analytics, news, AI assistant, watchlist and Explore features. Assumed to have basic familiarity with share tickers but no formal finance training, therefore everything must be explainable in plain language.

EquityLens deliberately has a single in-app role. System administration (account management, configuration, monitoring) is an operational responsibility carried out by the development team through the hosting platform's own tools; it is not a user-facing capability of the system and no administrative interface is in scope. One non-human actor starts work on its own: the **Scheduler** (a scheduled GitHub Actions workflow) runs the nightly news collection described in UC-26.

2.2 User Stories

2.2.1 Authentication and Account

- US-1: As a new user, I want to create an account so that my portfolio data is saved securely.
- US-2: As a new user, I want to verify my email address with a code so that only I can activate my account.
- US-3: As a security-conscious user, I want to protect my account with an authenticator app code so that a stolen password alone cannot access my data.
- US-4: As a returning user, I want to log in and stay logged in while I work.
- US-5: As a user who has forgotten my password, I want to reset it with a code sent to my email.
- US-6: As a user on a shared device, I want to log out so that others cannot access my data.
- US-7: As a user leaving the platform, I want to delete my account and all of my data permanently.

2.2.2 Portfolio Import and Management

- US-8: As an EasyEquities user, I want to upload my PDF statement so that the system extracts my holdings, trades, contributions, dividends and fees automatically.
- US-9: As an investor with another broker, I want to download an Excel template, fill it in and upload it.
- US-10: As an investor with more than one account, I want each statement kept as its own portfolio so that I can view each one's summary.
- US-11: As an investor, I want to download a PDF report of my portfolio so that I can keep or share a snapshot.

2.2.3 Dashboard and Insights

- US-12: As an investor, I want a single health score for my portfolio so that I can tell at a glance whether it is well spread.
- US-13: As an investor with a particular strategy, I want to choose how the health score is measured so that it matches my goals.

- US-14: As an investor, I want to see my performance charted against a relevant market benchmark so that I know if I am beating the market.
- US-15: As a busy user, I want the system to tell me what needs my attention (concentration risk, poor diversification) rather than making me hunt through data.
- US-16: As an investor, I want to see which holdings drove today's gain or loss so that I know why my portfolio moved.
- US-17: As an investor, I want to see what adding to a sector or rebalancing would do to my health score before I act.

2.2.4 Unusual Moves and News

- US-18: As an investor, I want to be shown the days on which my holdings moved unusually far so that I notice important events.
- US-19: As a novice, I want each unusual move explained in plain language, including how much of it was the market and what news was published around it.
- US-20: As an investor, I want to read recent news about the companies I hold.

2.2.5 Analytics and Financial Indicators

- US-21: As an investor, I want to see the Sharpe and Sortino ratios of each of my holdings so that I can judge whether its returns justify its risk.
- US-22: As a novice, I want a plain-language explanation of every indicator so that I can learn what the numbers mean.
- US-23: As a technical user, I want to see the RSI for my stocks so that I can identify overbought or oversold conditions.
- US-24: As a user, I want to choose which indicators I see for each holding.

2.2.6 AI Assistant

- US-25: As a novice, I want to ask "Is my portfolio diversified?" and get a natural-language answer grounded in my actual holdings.
- US-26: As a user looking at a dashboard card, I want to ask the assistant about it with one click.
- US-27: As a returning user, I want my previous AI conversations saved so that I can pick up where I left off.
- US-28: As a user, I want to rename or delete old conversations so that my chat history stays organised.
- US-29: As a user, I want to see and delete what the assistant remembers about me.

2.2.7 Watchlist and Explore

- US-30: As an investor, I want to add shares I do not yet own to a watchlist so that I can track them before buying.
- US-31: As a user, I want to remove a share from my watchlist when I am no longer interested.
- US-32: As an investor, I want suggestions of JSE shares that are similar to the ones I hold.

2.2.8 General

- US-33: As a user, I want to switch between light and dark themes.

- US-34: As a user sharing my screen, I want to hide rand values.
- US-35: As a new user, I want a help centre that answers common questions.

3. Use Cases and Use Case Diagram

3.1 Use Case Diagrams

The use case model is presented as one overview diagram followed by one diagram per group of subsystems.

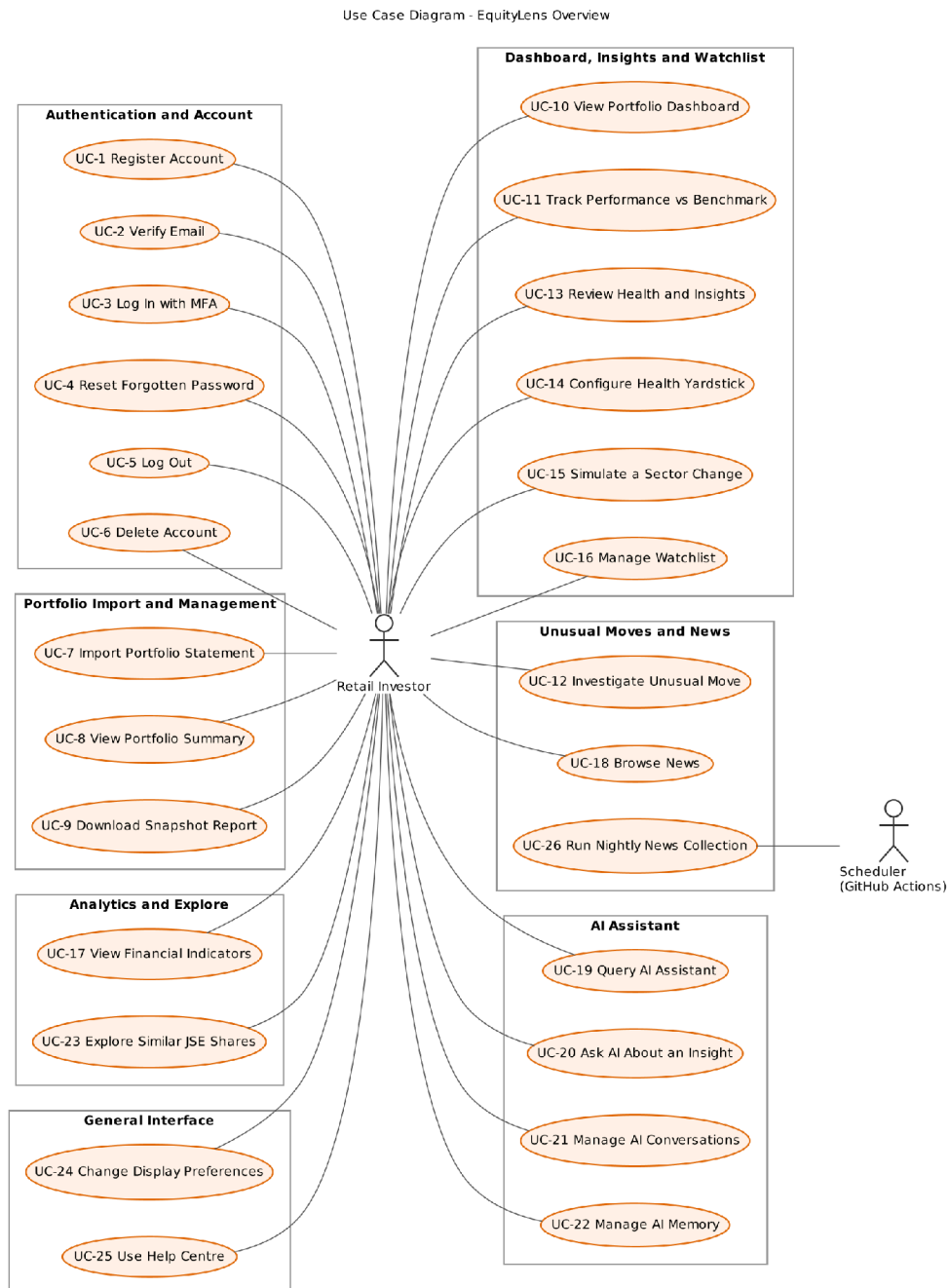


Figure 3.1 - Use case overview

Use Case Diagram - Authentication and Account

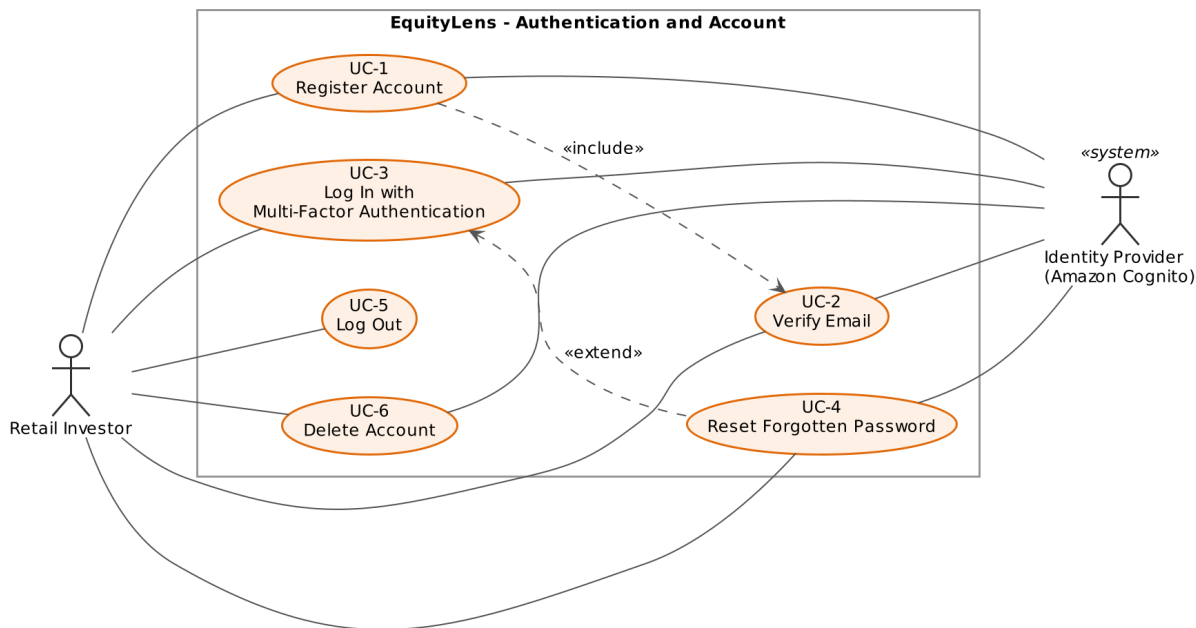


Figure 3.2 - Authentication and account
Use Case Diagram - Portfolio Import and Reports

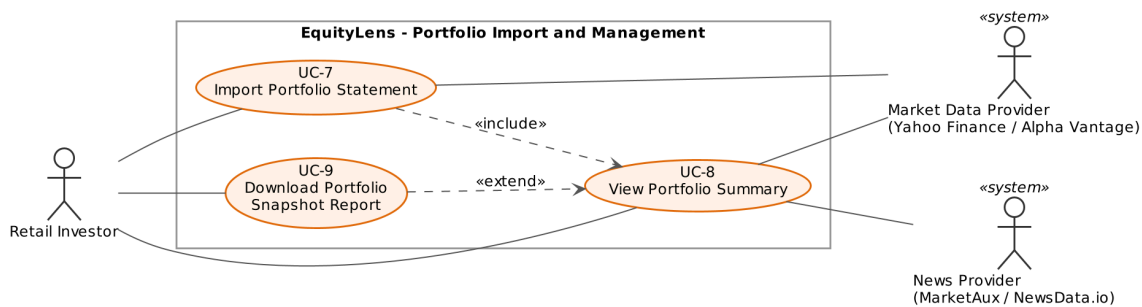


Figure 3.3 - Portfolio import and reports

Use Case Diagram - Dashboard and Watchlist

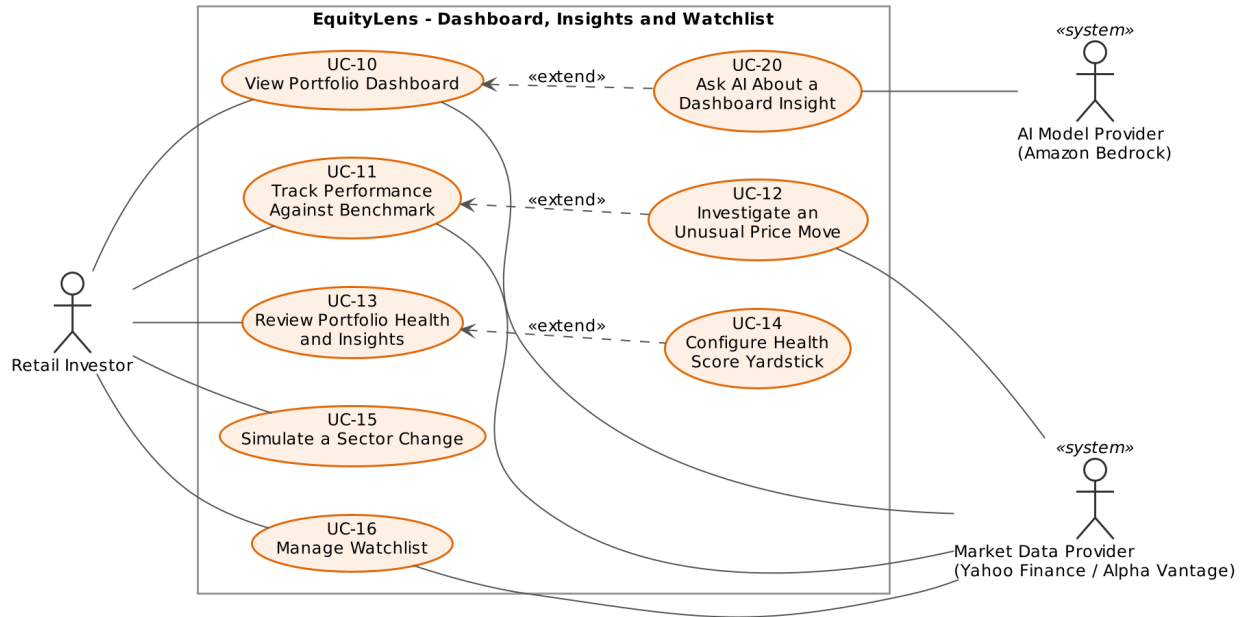


Figure 3.4 - Dashboard and watchlist

Use Case Diagram - Unusual Moves and News

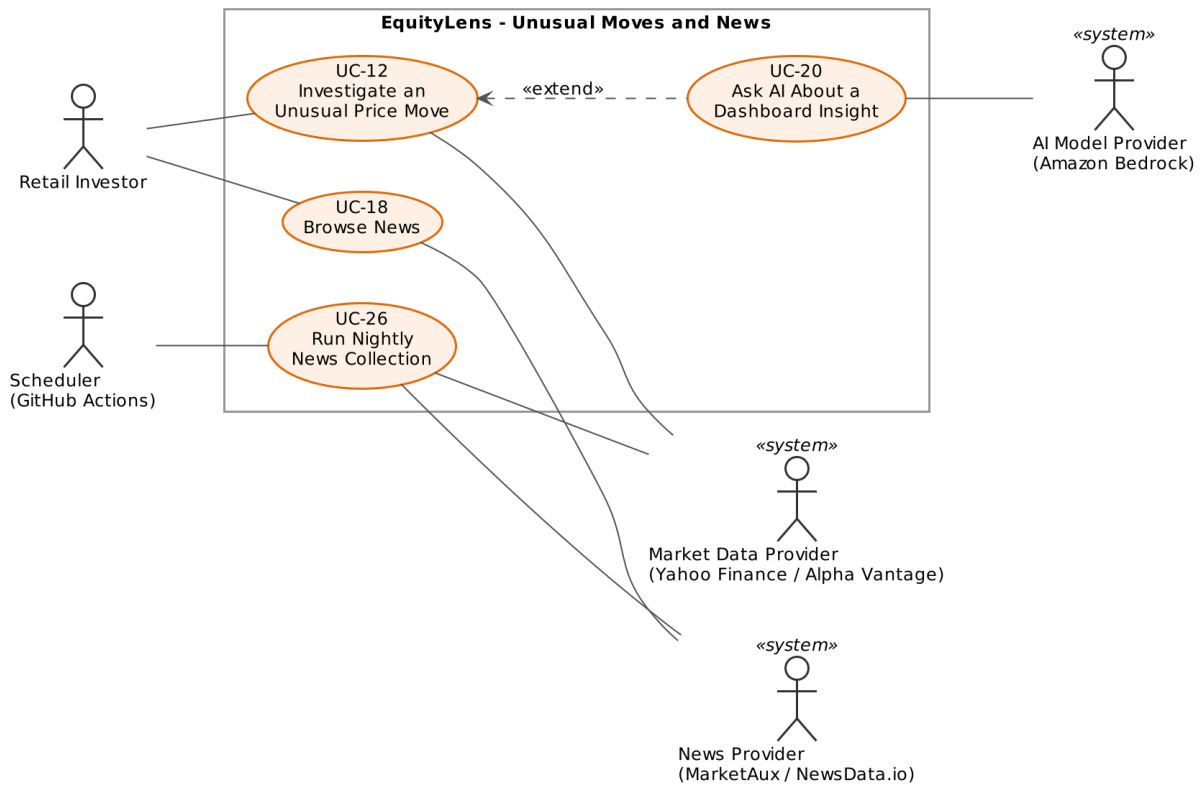


Figure 3.5 - Unusual moves and news

Use Case Diagram - Analytics, Explore and General Interface

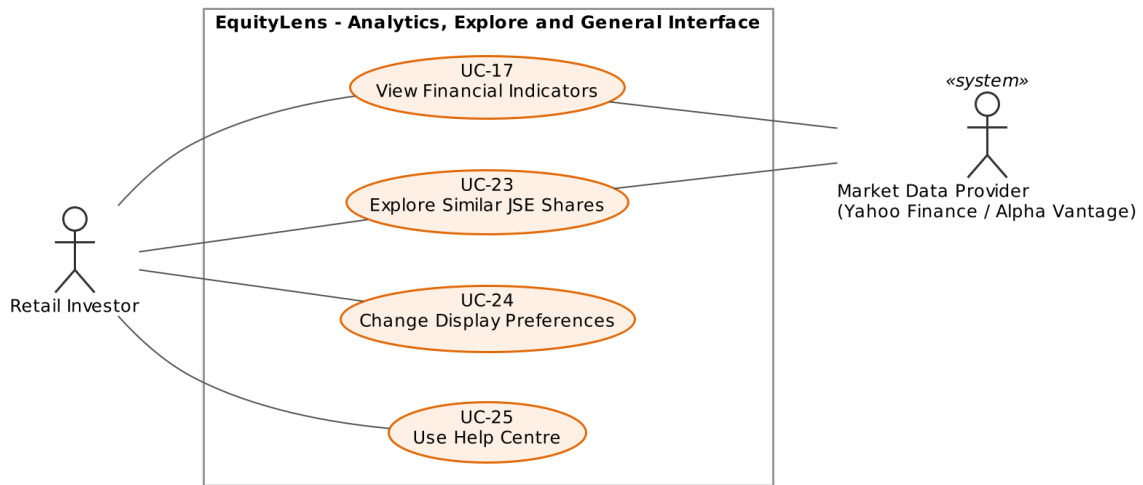


Figure 3.6 - Analytics, Explore and general interface

Use Case Diagram - AI Assistant

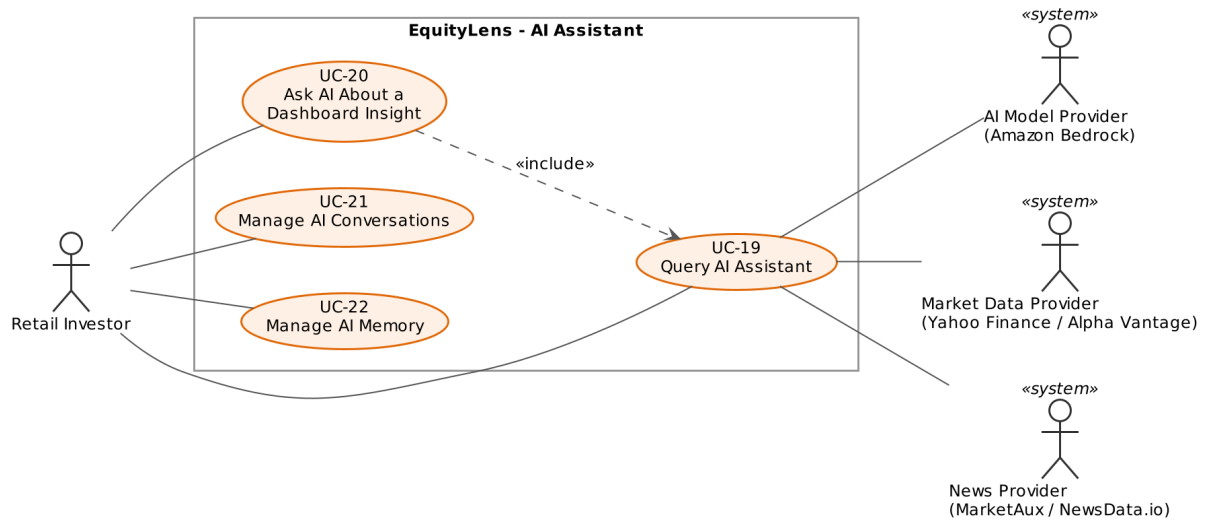


Figure 3.7 - AI assistant

3.2 Use Case Descriptions

UC-1: Register Account

Actors: Retail Investor (primary); Identity Provider (supporting)

Precondition: The user is not signed in and has opened the registration page (from Sign up or Analyse my portfolio on the landing page, or Create one on the login page).

Postcondition: An unconfirmed account exists at the identity provider and a verification code has been sent to the user's email address.

TUCBW the user opens the registration page ("Create your account").

Main flow:

1. The system displays the Full Name, Email Address, Password and Confirm Password fields.
2. The user completes the fields. The system validates each field when the user leaves it and shows any error directly beneath that field.
3. The user selects Create Account.
4. The system validates every field again and submits the registration to the identity provider.
5. The system shows "Account created. Check your email for a verification code." and, after 1.5 seconds, opens the Verify email page with the email address carried over (UC-2).

Alternative flow:

- Existing account: the user selects Sign in below the form and is taken to the login page.

Exception flows:

- Invalid input: a field fails validation (for example, a password without a special character). The system shows the message beneath that field and does not submit.
- Email already registered: the system shows "An account with this email already exists. Sign in instead."
- Other identity-provider error: the system shows the provider's message, or "Registration failed".

TUCEW the user is on the Verify email page, or has been shown why the registration was not accepted.

UC-2: Verify Email

Actors: Retail Investor (primary); Identity Provider (supporting)

Precondition: The user has registered and received a verification code by email.

Postcondition: The account is confirmed and the user can log in.

TUCBW the Verify email page opens after registration, or the user opens it directly.

Main flow:

1. The system shows "Verify email" and the address the code was sent to.

2. The user enters the 6-digit code in the Verification code field (characters other than digits are ignored).
3. The system enables Verify Email once exactly six digits have been entered, and the user selects it.
4. The system submits the code to the identity provider, which confirms the account.
5. The system shows "Email verified. Redirecting..." and opens the login page after 1.5 seconds.

Alternative flow:

- No email carried over: when the user reaches the page directly, the system also shows an Email Address field, which the user completes before submitting.

Exception flow:

- Invalid or expired code: the system shows the identity provider's error message, clears the code field and lets the user enter another code.

TUCEW the user is on the login page with a confirmed account.

UC-3: Log In with Multi-Factor Authentication

Actors: Retail Investor (primary); Identity Provider (supporting)

Precondition: The user has a confirmed account and is on the login page.

Postcondition: The user is authenticated and the dashboard is displayed.

TUCBW the user enters an email address and password and selects Sign In.

Main flow (authenticator already set up):

1. The system validates the fields and sends the credentials to the identity provider.
2. The identity provider accepts the password and asks for a TOTP code. The system shows "Two-factor authentication" and asks for the 6-digit code.
3. The user enters the code from their authenticator app and selects Verify (enabled once six digits have been entered).
4. The identity provider accepts the code and the system opens the dashboard.

Alternative flows:

- First login, authenticator not yet set up: at step 2 the identity provider asks for authenticator set-up. The system shows "Set up 2FA" with a QR code and an Enter key option that reveals the account name and secret key for manual entry. The user adds EquityLens to their authenticator app and selects Continue. The system shows "Verify your authenticator"; the user enters the current 6-digit code and selects Activate MFA (or Back to QR code). The identity provider activates the authenticator and completes sign-in, and the system opens the dashboard.
- No second factor requested: if the identity provider completes sign-in after the password, the system opens the dashboard directly.
- Forgotten password: the user selects Forgot password? (UC-4).

Exception flows:

- Incorrect credentials, or any other sign-in error: the system shows "Incorrect email or password" without saying which value was wrong.
- Invalid code: the system shows the identity provider's message, clears the code input and lets the user try again.

TUCEW the user sees their dashboard.

UC-4: Reset Forgotten Password

Actors: Retail Investor (primary); Identity Provider (supporting)

Precondition: The user has an account and is on the login page.

Postcondition: The user's password has been changed.

TUCBW the user selects Forgot password? on the login page.

Main flow:

1. The system shows "Forgot your password?" with an Email Address field.
2. The user enters their email address and selects Send Reset Code.
3. The system asks the identity provider to email a reset code, shows "If an account exists for email, a reset code has been sent. Redirecting..." and opens "Reset your password" after 1.5 seconds.
4. The user enters the 6-digit code, a new password and its confirmation. The system validates the new password with the registration rules.
5. The user selects Reset Password (enabled once the code has six digits).
6. The system shows "Password reset. Redirecting to sign in..." and opens the login page.

Exception flows:

- Too many attempts: the system shows "Too many attempts. Please wait a while before trying again."
- Email not yet verified: the system shows "This account has not verified its email yet - check your inbox for a verification code, or register again."
- Code rejected: the system shows the identity provider's message (or "Reset failed") and keeps the values entered.

TUCEW the user is on the login page and can sign in with the new password.

UC-5: Log Out

Actors: Retail Investor (primary)

Precondition: The user is signed in.

Postcondition: The session's tokens are removed from the browser and protected pages require a new login.

TUCBW the user selects Sign out in the top bar.

Main flow:

1. The system signs the user out and removes the stored tokens from the browser.
2. The system shows the login page (if the user was on the Help page, the public Help page is shown instead). Any other protected page now redirects to the login page until the user signs in again.

Alternative flow:

- Session no longer valid: when the server rejects the access token as expired or invalid, the system signs the user out automatically and opens the login page.

TUCEW the user is signed out and sees the login page (or the public Help page).

UC-6: Delete Account

Actors: Retail Investor (primary); Identity Provider (supporting)

Precondition: The user is signed in and on the Settings page.

Postcondition: The user's data and identity-provider account no longer exist.

TUCBW the user types their email address in the Danger Zone of the Settings page.

Main flow:

1. The system enables Delete Account once the typed text matches the signed-in email address (ignoring letter case and surrounding spaces).
2. The user selects Delete Account. The system lists what will be deleted and asks the user to Continue or Cancel.
3. The user selects Continue. The system asks "Are you absolutely sure? This is your final confirmation for email."
4. The user selects Permanently Delete.
5. The system deletes the user's portfolios, holdings, transactions, contributions, dividends, expenses, snapshots, uploaded-document records, watchlist, AI conversations, AI memories, preferences and user record, and then deletes the identity-provider account.
6. The system signs the user out and opens the landing page.

Alternative flow:

- Cancel: the user selects Cancel at step 2 or 3 and nothing is deleted.

Exception flow:

- Deletion fails: the system shows the server's message, for example "email does not match your account" or "Account deletion failed, contact support".

TUCEW the user is on the landing page and the account no longer exists.

UC-7: Import Portfolio Statement

Actors: Retail Investor (primary); Market Data Provider (supporting, for resolving instrument names)

Precondition: The user is signed in, is on the Portfolio page, and has an EasyEquities PDF statement or a completed EquityLens Excel template (.xlsx).

Postcondition: A new portfolio with its holdings and transaction records is saved against the user's account, its summary is displayed, and it becomes the portfolio shown on the Dashboard.

TUCBW the user chooses an Account Type in the "Upload your statement" card.

Main flow:

1. The user selects ZAR, TFSA or USD as the account type.
2. The user selects Choose File, and the system opens the file picker limited to .pdf and .xlsx files.
3. The user selects a PDF statement. The system asks for the statement's password ("Enter the password for your EasyEquities statement."); the user enters it, or leaves it blank, and selects Continue.
4. The system shows "Loading Portfolio" and, in the browser, extracts the account number, portfolio name, statement end date, holdings, purchases and sales, contributions and withdrawals, dividends and withholding tax, and expenses.

5. The system sends the extracted rows, not the file, to the server. The server records the file name, creates the portfolio, resolves each instrument name to a ticker and sector, and saves the rows.
6. The system displays the new portfolio's summary (UC-8) and refreshes My Portfolios.

Alternative flows:

- Excel template: at step 3 the user selects a completed template. The system reads its sheets in the browser (there is no password prompt) and continues at step 5.
- Download the template: before starting, the user selects Download Template in the Excel Template card (the template can also be downloaded from the Help page).
- Cancel: the user selects Cancel in the password dialog and nothing is imported.

Exception flows (each is shown in a "Something went wrong" dialog; the user selects OK and can choose another file):

- No account type selected: "Please select an account type first"
- Wrong PDF password: "Incorrect PDF password. Please check the password and try again"
- Unreadable PDF, or not an EasyEquities statement: "Could not read this PDF. Please make sure it is a valid EasyEquities statement."
- Unreadable Excel file: "Invalid or unsupported Excel file. Please make sure to use the Excel template"
- Saving fails: the system deletes the partly saved portfolio and shows "Could not save your statement: reason. Nothing was imported - please try again."

TUCEW the user sees the new portfolio's summary on the Portfolio page.

UC-8: View Portfolio Summary

Actors: Retail Investor (primary); Market Data Provider and News Provider (supporting)

Precondition: The user is signed in, is on the Portfolio page and has imported at least one portfolio.

Postcondition: The summary of the selected portfolio is displayed and a snapshot of it is stored, identified by its SHA-256 hash.

TUCBW the user selects View Summary on a portfolio.

Main flow:

1. The system lists the two most recently imported portfolios in My Portfolios, each with its name, account-type badge, account number and statement date.
2. The user selects View Summary on one of them.
3. The system builds a snapshot of the portfolio (including its analytics and current news) and stores it with a SHA-256 hash.
4. The system shows six summary cards (Portfolio Value at cost, Holdings, Purchase & Sales, Contributions, Dividends, Expenses), the Smart Portfolio Snapshot panel, and charts of trading activity, cash flow, dividend income per instrument, allocation, top holdings and the lowest holding.

Alternative flows:

- All portfolios: the user selects View all portfolios, and then View Summary on any portfolio in the list.

- Ask the assistant: the user selects Go To Assistant to open the AI Assistant page.

Exception flow:

- Load failure: the system shows "Could not load portfolio summary".

TUCEW the user sees the selected portfolio's summary.

UC-9: Download Portfolio Snapshot Report

Actors: Retail Investor (primary)

Precondition: A portfolio summary is displayed on the Portfolio page (UC-8, or after UC-7).

Postcondition: A PDF report of the portfolio is saved on the user's device.

TUCBW the user selects Download Portfolio Summary in the Smart Portfolio Snapshot panel.

Main flow:

1. The system generates a PDF from the stored snapshot and shows the snapshot's SHA-256 hash in it.
2. The report contains the portfolio summary, allocation, top holdings, trading activity, dividend income, cash flow, market news and the seven financial indicators. For the most recently imported portfolio it also contains What Moves Your Money, Portfolio at a Glance, Portfolio Health, Holdings and Tax (with a capital gains tax estimate), Risk Assessment, and Risk Drivers and Tax.
3. The browser saves the file as equity-lens-portfolio-id.pdf.

Exception flow:

- Download fails: the system shows "Could not download the portfolio Summary. Please try again later".

TUCEW the report has been saved.

UC-10: View Portfolio Dashboard

Actors: Retail Investor (primary); Market Data Provider (supporting)

Precondition: The user is signed in.

Postcondition: The dashboard reflects the current state of the user's most recently imported portfolio, and today's value snapshot is recorded.

TUCBW the user opens the Dashboard (after login, or from the sidebar).

Main flow:

1. The system confirms the session.
2. The system values each holding at its latest cached closing price, refreshing stale prices, converts US-dollar prices to rand, and records the portfolio's value for today.
3. The system shows, in order: Overview (greeting, a one-sentence summary of today's move, Portfolio Value, Today, Invested, Investment Gain, the health label and the difference against the benchmark), Portfolio Insights, Portfolio Health, Performance vs Benchmark, All Positions (with sector allocation and market context), Concentration & Rebalancing, and a Watchlist button.

Alternative flows:

- No holdings: each card shows its empty message, for example "Import a portfolio to see your figures." and "Upload holdings to see them here."

- Holding without a current price: the holding is valued at its statement price, or at cost, and its Today column shows "-".
- Filter by sector: the user selects a sector (or a row) and All Positions shows only that sector's holdings until the user selects Clear.

Exception flows:

- Unauthenticated user: the system redirects to the login page.
- Data retrieval failure: the system shows "Could Not Load Portfolio" with Try again and Go to Portfolio.
- One card fails: that card shows "This section could not be displayed." with Try again, and the other cards still render.

TUCEW the user sees their dashboard.

UC-11: Track Performance Against Benchmark

Actors: Retail Investor (primary); Market Data Provider (supporting)

Precondition: The dashboard is open and the portfolio has at least two days of value history, stored or reconstructed.

Postcondition: The user has seen the portfolio charted against its benchmark with comparative returns.

TUCBW the user views the Performance vs Benchmark card.

Main flow:

1. The system retrieves the portfolio's daily values. Where the stored history starts after the first imported transaction, it reconstructs the earlier days from the transactions and cached closing prices.
2. The system computes the benchmark: regional index proxies (Satrix 40 for holdings classified as South African, the S&P 500 through SPY for US, MSCI World through URTN for global and MSCI Emerging Markets through EEM for emerging markets), converted to rand, blended by each region's share of the user's classified holdings and rebased to the portfolio's starting value.
3. The system charts the portfolio and the benchmark and shows Portfolio return (time-weighted), the benchmark's return and the difference for the selected range, with a one-sentence takeaway.
4. The user selects a range (1D, 1W, 1M, 3M, 1Y or ALL) or zooms with Ctrl + scroll, and the figures update.

Alternative flows:

- Compare holdings: the user adds up to five individual holdings and up to four named groups of holdings (at most 15 distinct holdings in total), drawn as lines indexed to 100.
- History settings: the user selects the gear and chooses reconstructed history or only since upload, and how far back to go (3 months, 6 months, 1 year or everything).
- Unusual moves: the user selects a marker on the chart (UC-12).

Exception flows:

- Insufficient history: the chart area shows "Building history - N days so far" with an explanation.

- Benchmark closing price missing for a day: the last known benchmark level is carried forward; where there is none, the benchmark line has a gap.
- No holdings that can be assigned a region: the benchmark is shown as "No benchmark".

TUCEW the user sees their performance plotted against the benchmark with the comparative return figures.

UC-12: Investigate an Unusual Price Move

Actors: Retail Investor (primary); Market Data Provider (supporting)

Precondition: The dashboard is open, and at least one holding has 60 or more usable daily returns.

Postcondition: The user has seen the explanation of the selected move.

TUCBW the user selects a marker (clicks it, or presses Enter or Space on it) on the Performance vs Benchmark chart.

Main flow:

1. When the dashboard loads, the system scans each holding's daily returns and flags the days on which the move was at least three times the holding's own EWMA volatility. Up to eight markers are drawn on the portfolio line (red for unusual falls, green for unusual rises), with the caption "Each dot is a day one of your holdings moved unusually far for itself. Click one to see what it meant for you."
2. The user selects a marker, and the system opens the event card with the holding, the size of the move and the date.
3. For a holding in the built-in instrument catalogue, the system shows Impact on your portfolio: the holding's weight on the day before the move (or today's weight when that day's portfolio value is not known) multiplied by the move.
4. The system shows How unusual was this?: the band (Unusual, Very unusual or Extremely unusual), the move as a multiple of the holding's normal daily move and, for a holding in the built-in instrument catalogue, the split between the market's part and the company-specific part from a market-model event study against the holding's regional benchmark and, when enough other tracked large caps were scanned, whether several of them also moved unusually that day.
5. For a holding in the built-in instrument catalogue, the system shows What might explain it?: up to five stored articles tagged with the holding and published between three days before and one day after the move (exchange-local dates). Each has a link that opens in a new tab, a Closely related or Related label, its source, date and timing, and a highlighted sentence, followed by "These articles were published around the move and mention ticker. That doesn't mean they caused it."
6. The user closes the card with the X button.

Alternative flows:

- Explore possible causes: the system opens the assistant dock with a prepared question (UC-20).
- Detection & event-study details: the system shows the detector settings, beta, alpha, R squared, the estimation and event windows, the cumulative abnormal return chart with its 95% band, the working, the per-article scores and when news was last collected.

Exception flows:

- No article qualifies: the system shows "No matching news found" and explains that no stored article is close enough to the date to point to a possible cause.
- Holding outside the built-in instrument catalogue: the holding has no regional benchmark, so the event card shows the move, date and band but no impact, event study or articles, and shows "No matching news found".
- Detail cannot be loaded: the system shows "Could not load the detail for this move."
- No markers: the caption reads "No unusual moves across N holdings in the last year.", "Not enough price history yet to look for unusual moves." or "Unusual-move markers are unavailable right now."

TUCEW the user has read the explanation and closed the card.

UC-13: Review Portfolio Health and Insights

Actors: Retail Investor (primary)

Precondition: The dashboard is open.

Postcondition: The user has seen the health score, its factors and any items that need attention.

TUCBW the user views the Portfolio Health and Portfolio Insights cards, or selects Review Portfolio Health in the Overview.

Main flow:

1. The system computes a health score out of 10 as the weighted sum of three structural subscores (Sector Concentration, Single-Stock Risk - shown as Fund Concentration when the largest holding is a fund - and Portfolio Breadth) using the user's chosen yardstick.
2. The system shows the score, its label (Excellent, Healthy, Mixed or Needs attention) and each subscore with its weight, its value and a Why reveal giving the detail, the target and how to improve.
3. The system shows Portfolio Insights. Candidate cards come from today's biggest gainer and drag and from rule templates, for example: the largest holding or largest sector at or above the high-concentration threshold; the largest holding between the low (watch) and high thresholds; two or fewer positions; the top three holdings above 50% of the portfolio; effective positions below half the position count; two or fewer sectors; and the subscore costing the most points. There are also tax, performance, drawdown, volatility and data-quality observations. Cards are ranked by severity and size, and at most two cards are kept per category (three for today's movers). The first eight are shown, and the rest are behind "Show N more".
4. The user selects Why? on a card to see its explanation, evidence and an Ask AI Why button, or selects the card's action button, which scrolls to the relevant dashboard section (the stale-statement card's Import a statement button opens the Portfolio page).

Alternative flows:

- Ask AI Why: the system opens the assistant dock with the insight as a question (UC-20).
- Change the yardstick: the user selects the gear on Portfolio Health (UC-14).
- Nothing to report: Portfolio Insights shows "Nothing notable moved today."
- No holdings: Portfolio Health shows "Health score appears once you have holdings to analyse."

TUCEW the user knows their health score and what, if anything, needs their attention.

UC-14: Configure Health Score Yardstick

Actors: Retail Investor (primary)

Precondition: The dashboard is open and the portfolio has holdings, so the Portfolio Health card shows a score.

Postcondition: The chosen yardstick is saved against the user's account and applied to the dashboard.

TUCBW the user selects the gear icon on the Portfolio Health card.

Main flow:

1. The system opens "Portfolio health scoring", showing the yardstick in use and eight presets: EquityLens default, Capital preservation, Income / dividend, Balanced, Growth, Institutional (5/10/40), Core / satellite, and Concentrated / high conviction.
2. The user selects a preset.
3. The system saves the choice and recalculates the dashboard.

Alternative flows:

- Reset to EquityLens: the system restores the default yardstick.
- Advanced - define your own: the user edits the sector spread weight, single-position weight, breadth weight, the two concentration thresholds, the sector HHI target and the effective-positions target, and selects Save my settings once Weights total shows 100%.

Exception flows:

- Value out of bounds: each weight must be 5-70% with the three summing to 100%, the thresholds 10-70% with the lower below the upper, the HHI target 0.05-0.50 and the breadth target 3-20. The system shows the server's message and nothing changes.
- Other save failure: the system shows the server's message if it sends one (for example "Something went wrong"), otherwise "Couldn't save that. Settings are unchanged."

TUCEW the dashboard reflects the chosen yardstick.

UC-15: Simulate a Sector Change

Actors: Retail Investor (primary)

Precondition: The dashboard is open and the portfolio spans at least two sectors.

Postcondition: The user has seen the simulated effect on the health score. No portfolio data has changed.

TUCBW the user uses the Concentration & Rebalancing card.

Main flow (add to a sector):

1. The system proposes the smallest sector, and the user may choose another.
2. The user selects Simulate investing in sector.
3. The system adds 5% of the portfolio's value to that sector's holdings in proportion to their size, recalculates the health score, and shows the score before and after, the subscore changes and the illustrative amount, with "Analysis only, not a trade instruction - EquityLens doesn't execute trades."

Alternative flows:

- Rebalance: the user selects Simulate shifting largest into smallest. If the largest sector is at or above the high-concentration threshold, the system moves the excess down to the low threshold into the smallest sector and shows the change in health score, a Now and Simulated allocation bar and the steps.
- Clear: the user removes a result.

Exception flows:

- Fewer than two sectors: "Sector what-ifs need at least two sectors to compare"
- Nothing over the threshold: "No sector is over the high% concentration threshold right now - nothing to rebalance."
- Simulation fails: "Couldn't run this simulation right now."

TUCEW the user has seen the simulated result.

UC-16: Manage Watchlist

Actors: Retail Investor (primary); Market Data Provider (supporting)

Precondition: The user is signed in and on the Dashboard.

Postcondition: The user's watchlist reflects the additions or removals made.

TUCBW the user selects the Watchlist (star) button on the Dashboard.

Main flow (add):

1. The system opens the watchlist panel, and the user selects + Add.
2. The user types at least two characters of a ticker or company name.
3. The system searches the market data provider and lists up to ten matching shares, ETFs or indices, each with its symbol and name.
4. The user selects a match, or keeps the ticker typed, and selects Add.
5. The system saves the entry and shows the first three entries of the watchlist, each with its ticker, company name (or the ticker if none is known), the latest price prefixed "R" (JSE prices are converted from cents to rand; other listings are shown in their own currency) and daily percentage change.

Alternative flow (remove):

- The user selects the X on an entry, and the system deletes it and refreshes the list.

Exception flows:

- No match: the list shows "No matches"; if the search fails it shows "Search failed".
- Instrument that cannot be priced: the system shows the server's reason, for example "ticker is a type, not a stock, ETF, or index, so it can't be priced or displayed here." (otherwise "Could not add that ticker.")
- List cannot be loaded: "Couldn't load your watchlist."

TUCEW the user sees their updated watchlist.

UC-17: View Financial Indicators

Actors: Retail Investor (primary); Market Data Provider (supporting)

Precondition: The user is signed in and holds at least one instrument with a ticker.

Postcondition: The indicators for each holding are displayed.

TUCBW the user opens Analytics from the sidebar.

Main flow:

1. The system collects the distinct tickers held across all of the user's portfolios.
2. For each ticker, the system calculates the CAPM expected return, P/E ratio, Altman Z-score, Sharpe ratio, Beta, Sortino ratio and RSI from cached prices and company fundamentals.
3. The system shows one card per holding, with a cell per indicator showing the value to two decimal places, a short classification and a colour (green favourable, red unfavourable).
4. The user hovers over an indicator name to see a short explanation, or selects a value to open a dialog with Why it matters, How it's worked out, the formula, any sector note and the values across the holdings that have one.

Alternative flows:

- Customise: the user selects Edit on a holding, picks a preset (Most Popular, Value Investing, Risk & Volatility, Technical / Momentum or All Indicators) or individual indicators, optionally saves the selection as a named preset, and applies it to that holding or to all holdings. The choice is remembered in the browser.
- Ask the assistant: the user selects "Still not understanding the indicators? Ask the AI Assistant", which opens the AI Assistant in a new tab.

Exception flows:

- Indicator cannot be calculated (for example P/E for a fund, or Altman Z-score for a bank): the cell shows N/A with a short reason, and the other cells are unaffected.
- No price history for a holding: that holding's cells show Error and "Calc failed".
- No holdings: "No Holdings" with an Import Holdings button.
- Request fails: "Failed to load indicators".

TUCEW the user sees the indicator cards for their holdings.

UC-18: Browse News

Actors: Retail Investor (primary); News Provider (supporting)

Precondition: The user is signed in.

Postcondition: News relevant to the user's holdings, or general market news, is displayed.

TUCBW the user opens News from the sidebar.

Main flow:

1. The system retrieves the tickers held across the user's portfolios.
2. For each ticker, the system refreshes the stored news from the news provider if it has not done so in the last six hours, and returns up to 20 stored articles tagged with that ticker, newest first.
3. The system shows counters (Relevant Articles, Positive Impact, Negative Impact, Neutral Impact) and each article with its image, headline, description, source, date, ticker badge and a sentiment label.
4. The user selects Read article to open the source article in a new tab.

Alternative flows:

- Filter by ticker: the user selects a ticker chip; All shows every ticker again.
- General market news: the user selects the All Market tab and a category (Business, Top, Technology, Politics or Crime), and the system shows the latest market news for that category.

Exception flows:

- No holdings: "No holdings found" with an Import Holdings button.
- No articles: "No news available" and "There are currently no recent articles for your holdings."
- News provider unavailable: the previously stored articles, or the empty state, are shown.

TUCEW the user sees the news list for their selection.

UC-19: Query AI Assistant

Actors: Retail Investor (primary); AI Model Provider (supporting); Market Data Provider and News Provider (supporting, through the assistant's tools)

Precondition: The user is signed in and is on the AI Assistant page, or has opened the assistant dock on another page.

Postcondition: The assistant's response is displayed and the exchange is saved in the active conversation.

TUCBW the user types a question and presses Enter or selects Send.

Main flow:

1. The system sends the question to the AI model together with the facts it remembers about the user, the recent turns of the conversation (older turns that no longer fit are condensed into a running summary, which is also sent), and the portfolio context (up to the 15 largest holdings per portfolio, the health score and subscores, and the names of uploaded documents).
2. When needed, the model calls tools that fetch share prices, indicators, market news, statement detail, ticker matches or a goal projection.
3. The system streams the answer beneath the question as formatted text. A persistent disclaimer reads "AI responses are informational only and not financial advice."
4. The system saves the question and answer, gives a new conversation a short title, and stores any lasting facts about the user's goals and preferences in memory.

Alternative flows:

- Portfolio scope: before asking, the user selects the + button (Chat about) and chooses All portfolios or one portfolio.
- Suggested prompt: in an empty conversation, the user selects one of three suggested prompts, which is sent immediately.
- Copy or Regenerate: the user copies an answer, or regenerates the last answer.
- Assistant dock: on any other signed-in page, the user opens the assistant dock and asks there; the dock uses the same active conversation.

Exception flows:

- Answer cannot be generated: the system shows "Something went wrong, try again." with Retry on the AI Assistant page.

- Rate limit: after 5 messages in 60 seconds, or 50 in a UTC day, the system shows the server's message and a countdown.
- Request for a buy or sell recommendation: the model is instructed to explain the trade-offs and factors rather than tell the user to buy or sell a specific security.

TUCEW the user sees the assistant's response in the conversation.

UC-20: Ask AI About a Dashboard Insight

Actors: Retail Investor (primary); AI Model Provider (supporting)

Precondition: The user is on the Dashboard.

Postcondition: The assistant dock shows the question and its answer.

TUCBW the user selects EquityLens Insight on a dashboard card (Portfolio Health, Performance vs Benchmark, Concentration & Rebalancing or All Positions), Ask AI Why on an insight, or Explore possible causes on an event card.

Main flow:

1. When a card offers several suggested questions, the system lists them and the user selects one.
2. The system opens the EquityLens Assistant dock and sends the question (UC-19).
3. The answer streams into the dock beneath "Answers are AI-generated and can be wrong. Informational only, not financial advice."
4. The user asks follow-up questions, or closes the dock with the X button or Escape.

Exception flow:

- Answer cannot be generated: the dock shows "Something went wrong, try again."

TUCEW the user has read the answer in the dock.

UC-21: Manage AI Conversations

Actors: Retail Investor (primary)

Precondition: The user is signed in and on the AI Assistant page.

Postcondition: The conversation list reflects the changes made, and the selected conversation's history is displayed.

TUCBW the user interacts with the Conversations list on the AI Assistant page.

Main flow (resume a conversation):

1. The system lists the user's saved conversations, most recently active first, each with its title and time.
2. The user selects a conversation.
3. The system loads and displays its full message history and restores the portfolio scope last used with it in this browser (otherwise All portfolios).
4. The user continues the conversation.

Alternative flows:

- New conversation: the user selects New Chat. The system clears the message area and resets the portfolio scope, and the next question starts a new saved conversation.

- **Rename:** the user selects the pencil icon, edits the title and presses Enter to save; selecting elsewhere cancels.
- **Delete:** the user selects the trash icon. The system removes the conversation immediately, and if it was open, returns the message area to the greeting.

Exception flow:

- **List unavailable:** the list shows "No saved chats yet", and the user can still start a new conversation.

TUCEW the user sees the conversation list and message area reflecting their action.

UC-22: Manage AI Memory

Actors: Retail Investor (primary)

Precondition: The user is signed in and on the AI Assistant page.

Postcondition: The facts the assistant remembers reflect the deletions made.

TUCBW the user selects Memory at the bottom of the Conversations list.

Main flow:

1. The system opens "What it remembers about you", listing the stored facts oldest first, with the note "These carry across all your chats. Deleting one means the assistant forgets it."
2. The user selects the trash icon next to a fact.
3. The system deletes the fact, and the assistant no longer receives it.
4. The user closes the dialog.

Alternative flow:

- **Nothing stored:** the dialog shows "Nothing remembered yet. Tell the assistant about your goals and it will keep track of them between chats."

TUCEW the dialog is closed and the memory reflects the user's changes.

UC-23: Explore Similar JSE Shares

Actors: Retail Investor (primary); Market Data Provider (supporting)

Precondition: The user is signed in.

Postcondition: Suggestions of similar JSE shares are displayed.

TUCBW the user opens Explore from the sidebar.

Main flow:

1. The system collects the JSE-listed shares the user holds across their portfolios.
2. The system compares each of them with a universe of up to 120 of the largest JSE shares (market capitalisation of at least R1 billion) on size, free float and dividend yield, with a penalty when the sector differs.
3. The system shows up to nine shares the user does not already hold, each with a description such as "n% match to ticker: same sector, similar traits".
4. Below the suggestions, under "See it on the map", the system plots the holdings and suggestions by market capitalisation and free float, sized by dividend yield and coloured by sector. The user may select a holding or a suggestion to highlight how it connects to the others.

Alternative flow:

- Holdings that are not JSE shares: they are listed as not included, with "Explore only covers JSE-listed shares."

Exception flows:

- No JSE-listed shares held: "Explore finds JSE shares similar to the ones you hold. Your portfolio doesn't contain any JSE-listed shares yet."
- No holdings: "Explore isn't available for this portfolio"
- Market data unavailable: "Couldn't load recommendations: market data is temporarily unavailable. Try again shortly."

TUCEW the user sees the suggestions and map.

UC-24: Change Display Preferences

Actors: Retail Investor (primary)

Precondition: The user is signed in.

Postcondition: The chosen preference is applied and remembered in this browser.

TUCBW the user selects the theme switch under Appearance on the Settings page, or the blur button in the top bar.

Main flow (theme):

1. The user selects the theme switch.
2. The system switches between the light and dark themes immediately.
3. The system stores the choice in the browser and applies it before the page is drawn on later visits.

Alternative flow (hide rand values):

- The user selects the blur button in the top bar. The system blurs the rand figures in the dashboard's Overview, the All Positions table, the performance and contribution charts and the watchlist (useful while screen-sharing) and remembers the choice; selecting the button again shows them.

TUCEW the display reflects the user's preference.

UC-25: Use Help Centre

Actors: Retail Investor (primary; signed in or not)

Precondition: None.

Postcondition: The user has found guidance, an answer or a resource.

TUCBW the user opens Help from the sidebar, or Help Centre from the landing page.

Main flow:

1. The system shows the Help Centre with quick links (Getting Started, Import data, Understand your portfolio, News & market, AI Assistant), a quick tip, frequently asked questions and resources.
2. The user expands a question (Which file formats can I upload?, Can the AI Assistant see my portfolio?, Is this financial advice?) to read its answer, or selects a quick link.

Alternative flows:

- Resources: the user downloads the Portfolio template or selects Contact support to email the team.
- Signed out: the page is shown without the app's navigation, and the quick links lead to the login page.

TUCEW the user has read the help content or left through a link.

UC-26: Run Nightly News Collection

Actors: Scheduler (primary); Market Data Provider and News Provider (supporting)

Precondition: The production backend is running and a news provider key is configured.

Postcondition: Prices are refreshed, the unusual-move register is updated, new articles are stored and linked to tickers, and the run is recorded.

TUCBW the Scheduler starts the nightly job at 00:30 UTC (02:30 SAST).

Main flow:

1. The system takes a database lock so that only one run proceeds at a time.
2. The system refreshes daily prices for every held ticker except holdings whose name identifies them as a fund, for a fixed seed universe of 40 JSE and 10 US large caps, and for the regional benchmark proxies.
3. The system calculates its request allowance: at most 60 news requests for the run and at most 80 in the UTC day.
4. The system detects unusual moves across the tickers, validates them (rejecting price gaps, cents-to-rand unit flips and one-day spikes that reverse) and records them in the price-anomaly register.
5. Within the allowance, the system fetches the latest news published since the start of the last nightly run that finished ok or partial (or the last 36 hours if there is none), in batches of five tickers per country, held tickers first.
6. While allowance remains, for each validated unusual move in the last seven trading days that has no stored news evidence yet and has not already been fetched, the system fetches news published from four days before to two days after the move.
7. While allowance remains, the system fetches news for older validated moves from the last 183 days that have not yet been fetched.
8. The system stores new articles, links each one to the tickers the provider tagged as the same listed equity, records the run and prints a summary.

Alternative flows:

- Run already in progress: the job ends as skipped.
- Manual run: a team member starts the workflow in plan, dry-run or run mode.

Exception flows:

- Quota exhausted, repeated rate limiting or three consecutive failures: the run stops as partial and the workflow reports a warning.
- Provider rejects the key: the run stops as failed and the workflow fails.

TUCEW the run is recorded with its status.

4. Functional Requirements

Functional requirements are stated in terms of what the user can do and what the system visibly provides.

4.1 Subsystem 1: Authentication and Account

FR-1: User Registration

- FR-1.1: The system shall allow a new user to register with their full name, email address and a password entered twice (Password and Confirm Password).
- FR-1.2: The system shall check, before the registration is submitted, that the email address has the form name@domain.tld with a final part of two or three letters.
- FR-1.3: The system shall reject a password shorter than 8 characters or one that lacks an uppercase letter, a lowercase letter, a digit, or one of the special characters @ \$! % * ? & # . (full stop).
- FR-1.4: The system shall display an error message directly beneath each invalid or missing registration field when the user leaves the field and when the form is submitted.
- FR-1.5: The system shall reject a registration that uses an email address already associated with an account, and shall show "An account with this email already exists. Sign in instead."
- FR-1.6: After a successful registration, the system shall tell the user to check their email for a verification code and shall open the Verify email page with the email address carried over.

FR-2: Email Verification

- FR-2.1: The system shall send a verification code to the user's email address on registration, through the identity provider.
- FR-2.2: The system shall accept only a 6-digit numeric code on the Verify email page, and shall ask for the email address when it was not carried over from registration.
- FR-2.3: The system shall show the identity provider's error message when a code is rejected, clear the code field and allow the user to enter another code.
- FR-2.4: The system shall show "Email verified. Redirecting..." and open the login page once the email is verified.

FR-3: Login and Session

- FR-3.1: The system shall authenticate users by email address and password.
- FR-3.2: The system shall show "Incorrect email or password" for any failed sign-in, without revealing whether the email or the password was wrong.
- FR-3.3: The system shall keep the user signed in across pages and page reloads, refreshing access tokens silently while the session remains valid.
- FR-3.4: The system shall redirect an unauthenticated user who opens a protected page (Dashboard, Portfolio, Analytics, News, AI Assistant, Explore or Settings) to the login page.
- FR-3.5: The system shall redirect an authenticated user who opens the login, registration, email verification, forgot-password or reset-password page to the dashboard.

- FR-3.6: The system shall allow the user to sign out from the top bar; after signing out, protected pages shall redirect to the login page until the user signs in again.
- FR-3.7: When a request made through the application's API client is rejected with HTTP 401 because the access token is expired or invalid, the system shall sign the user out and open the login page. The Explore page and streamed AI answers instead show their own error message.
- FR-3.8: The system shall make the landing page, the Help Centre and the brand style page available without signing in.

FR-4: Multi-Factor Authentication

- FR-4.1: The system shall support a time-based one-time password (TOTP) from an authenticator app as the second authentication factor, requested after the password is accepted.
- FR-4.2: When the identity provider requires the user to set up an authenticator, the system shall show a QR code and, on request, the account name and secret key for manual entry.
- FR-4.3: The system shall require the user to confirm the pairing by entering a code generated by the authenticator app (Activate MFA) before sign-in completes.
- FR-4.4: On later sign-ins, the system shall ask for the current 6-digit code after the password is accepted.
- FR-4.5: The system shall reject an incorrect code with the identity provider's message, clear the code input and allow the user to try again.

FR-5: Password Reset

- FR-5.1: The system shall allow a user to request a password reset code by email from the Forgot password page, and shall respond with the same message whether or not an account exists for the address entered.
- FR-5.2: The system shall tell the user to wait when the identity provider reports too many attempts.
- FR-5.3: The system shall allow the user to set a new password by entering the 6-digit reset code, a new password that meets FR-1.3 and its confirmation.
- FR-5.4: After a successful reset, the system shall open the login page.

FR-6: Account Deletion

- FR-6.1: The system shall allow a signed-in user to delete their account from the Danger Zone on the Settings page, enabling the action only when the user has typed their own email address.
- FR-6.2: The system shall ask for two confirmations before deleting, listing what will be deleted.
- FR-6.3: On deletion, the system shall delete the user's portfolios, holdings, transactions, contributions and withdrawals, dividends, expenses, snapshots, uploaded-document records, watchlist, AI conversations and messages, AI memories, preferences and user record, and then delete the user's identity-provider account.
- FR-6.4: After deletion, the system shall sign the user out and open the landing page.

4.2 Subsystem 2: Portfolio Import and Management

FR-7: Statement Import

- FR-7.1: The system shall require the user to choose an account type (ZAR, TFSA or USD) before choosing a file to import.
- FR-7.2: The system shall accept two file types: an EasyEquities PDF statement (.pdf) and the EquityLens Excel template (.xlsx).
- FR-7.3: For a PDF statement, the system shall ask for the statement's password (which may be left blank) and shall extract the account number, portfolio name, statement end date, holdings (quantity, total cost, statement price and statement value; the statement's cost price and weight are used only to check each row, and the saved cost price and weight are recalculated from total cost), instrument purchases and sales, contributions and withdrawals, dividends and withholding tax, and expenses.
- FR-7.4: For the Excel template, the system shall read the sheets Portfolio, Holdings, Purchase and Sales, Contributions and Withdrawals, Dividends and Withholding Tax, and Expenses.
- FR-7.5: The system shall provide the Excel template for download on the Portfolio page and the Help page.
- FR-7.6: The system shall parse the statement file in the browser and send only the extracted rows and the file name to the server; the file itself shall not be uploaded or stored.
- FR-7.7: The system shall save the parsed data as a new portfolio immediately after parsing and then display that portfolio's summary.
- FR-7.8: The system shall report an unreadable file, a wrong PDF password or a missing account type in an error dialog, after which the user can choose another file.
- FR-7.9: If any step of saving or reloading the imported portfolio fails, the system shall delete the new portfolio and tell the user that nothing was imported.
- FR-7.10: The system shall keep each imported statement as a separate portfolio, so that a user may hold several portfolios.

FR-8: Instrument Resolution

- FR-8.1: The system shall resolve each imported instrument name to a ticker and sector using a built-in catalogue of nine JSE shares and eleven JSE-listed ETFs, ignoring letter case, punctuation and common fund words.
- FR-8.2: When a name is not in the catalogue, the system shall use the first result of a market data provider search.
- FR-8.3: When no ticker can be found, the system shall save the holding without a ticker and display it by its instrument name.

FR-9: Portfolio Summary

- FR-9.1: The system shall list the user's two most recently imported portfolios on the Portfolio page, each with its name, account-type badge, account number and statement date, and shall list all portfolios on request.
- FR-9.2: The system shall display, for a selected portfolio, the total cost of its holdings (labelled Portfolio Value), the number of holdings, and the totals of purchases and sales, contributions, net dividends and expenses.

- FR-9.3: The system shall chart, for a selected portfolio, trading activity, cash flow, dividend income per instrument (gross, withholding tax and net), allocation by weight, the top holdings and the lowest holding.

FR-10: Portfolio Snapshot Report

- FR-10.1: When a portfolio summary is displayed, the system shall build a snapshot of that portfolio (summary, charts, current news and analytics) and store it with its SHA-256 hash.
- FR-10.2: The system shall allow the user to download the stored snapshot as a PDF report that shows the hash, the portfolio summary, allocation, top holdings, trading activity, dividend income, cash flow, market news and the seven financial indicators.
- FR-10.3: For the user's most recently imported portfolio, the report shall also include What Moves Your Money, Portfolio at a Glance, Portfolio Health, Holdings and Tax (with a capital gains tax estimate), Risk Assessment, and Risk Drivers and Tax.

FR-11: Data Ownership

- FR-11.1: The system shall link every imported portfolio to the account of the user who imported it.
- FR-11.2: The system shall filter every read of portfolio, transaction, snapshot, watchlist and conversation data by the signed-in user, and shall answer a read request for another user's portfolio or conversation with "not found". News articles are stored once for all users, and a user can request news only for tickers they hold.

4.3 Subsystem 3: Dashboard and Insights

FR-12: Dashboard Overview

- FR-12.1: The system shall base the dashboard on the user's most recently imported portfolio.
- FR-12.2: The system shall value each holding at its latest cached closing price, falling back to its statement price and then to its cost when no price is available, and shall convert US-dollar prices to rand.
- FR-12.3: The system shall display a greeting, a one-sentence summary of today's move, the portfolio value, today's change in rand and percent, the amount invested in current holdings, the unrealised investment gain in rand and percent, the time of the last update, the health label and the return against the benchmark.
- FR-12.4: The system shall display each dashboard card independently, so that a card that fails shows "This section could not be displayed." with Try again while the other cards still render.
- FR-12.5: When the portfolio cannot be loaded, the system shall show "Could Not Load Portfolio" with Try again and Go to Portfolio; when the user has no holdings, each card shall show its own empty message.

FR-13: Portfolio Insights

- FR-13.1: The system shall rank insight cards by severity (risk, opportunity, neutral, information) and then by size, keep at most three daily-move cards and at most two cards from each other category (discarding the rest), show the first eight, and place the remainder behind "Show N more".

- FR-13.2: The system shall identify today's biggest gainer, today's biggest drag, and the holdings or sector behind most of today's move.
- FR-13.3: The system shall raise an insight when the largest holding or the largest sector reaches the yardstick's high-concentration threshold (45% by default), when the largest holding lies between the low and high thresholds (25-45% by default), when the portfolio has two or fewer positions, when the top three holdings exceed 50% of the portfolio, when the effective number of positions is below half the actual count, when the portfolio spans two or fewer sectors, and to name the health subscore costing the most points, subject to the per-category limit in FR-13.1.
- FR-13.4: For a ZAR account whose holdings can all be priced, the system shall show an estimate of the taxable capital gain that selling everything would produce (first R50,000 excluded, 40% inclusion, average cost) when it is at least R5,000; for a TFSA account it shall show the net contributions (contributions less withdrawals) when they are positive.
- FR-13.5: Each insight shall have a Why? reveal showing its explanation, its evidence and an Ask AI Why button; where the insight relates to a dashboard section, it shall also have a button that scrolls to that section.
- FR-13.6: When no insight applies, the system shall show "Nothing notable moved today."

FR-14: Portfolio Health

- FR-14.1: The system shall compute a structural health score from 0 to 10 as the weighted sum of three subscores: Sector Concentration (from the HHI of sector weights), Single-Stock Risk (from the largest holding's weight; shown as Fund Concentration when that holding is a fund) and Portfolio Breadth (effective number of positions, 1 divided by the sum of squared weights). Performance shall not affect the score.
- FR-14.2: The system shall label the score Excellent (8.5 or more), Healthy (7 or more), Mixed (5 or more) or Needs attention (below 5).
- FR-14.3: The system shall display each subscore with its weight, its value, and a plain-language detail, target and way to improve.
- FR-14.4: The system shall provide eight yardstick presets (EquityLens default, Capital preservation, Income / dividend, Balanced, Growth, Institutional (5/10/40), Core / satellite and Concentrated / high conviction), each defining the three weights, the low and high concentration thresholds, the sector HHI target and the effective-positions target. The EquityLens default uses weights of 40%, 35% and 25%, thresholds of 25% and 45%, an HHI target of 0.15 and a target of 8 effective positions.
- FR-14.5: The system shall allow the user to define their own yardstick, rejecting values outside the bounds of 5-70% for each weight (with the three summing to 100%), 10-70% for the thresholds (with the low below the high), 0.05-0.50 for the HHI target and 3-20 for the effective-positions target.
- FR-14.6: The system shall save the chosen yardstick against the user's account and apply it to the health score, the insights and the simulations.

FR-15: Performance vs Benchmark

- FR-15.1: The system shall record one value snapshot per day for the user's most recently imported portfolio when the dashboard is loaded, and shall reconstruct that portfolio's earlier daily values from its imported transactions and cached closing prices when at least 80% of its value can be priced.

- FR-15.2: The system shall compute the benchmark as a blend of regional index proxies (Satix 40 for South Africa, SPY for the S&P 500, URTX for MSCI World and EEM for MSCI Emerging Markets), converted to rand at each day's exchange rate, weighted by each region's share of the user's classified holdings and rebased to the portfolio's value on the first day of the series.
- FR-15.3: The system shall chart the portfolio's value and the benchmark together over a selectable range (1D, 1W, 1M, 3M, 1Y or ALL, with Ctrl + scroll zoom), and shall display the portfolio's return (time-weighted where the history allows), the benchmark's return, the difference, and a one-sentence takeaway for the range shown.
- FR-15.4: For days without a benchmark closing price, the system shall carry forward the last known level, and shall leave a gap only where no earlier level exists.
- FR-15.5: The system shall allow the user to compare up to five holdings, or up to four named groups of holdings, as lines indexed to 100.
- FR-15.6: The system shall allow the user to choose between reconstructed history and history since upload, and how far back the chart goes (3 months, 6 months, 1 year or everything), remembering the choice in the browser.
- FR-15.7: The system shall chart what the user put in against what it is worth now.
- FR-15.8: When fewer than two days of history exist, the system shall explain that the history is still being built.

FR-16: Positions and Sectors

- FR-16.1: The system shall list every holding sorted by value, showing its ticker and name, a concentration-risk badge (Low, Moderate or High against the yardstick's thresholds), its weight, today's percentage change and its value in rand, with an expandable row showing average cost, total return in rand and percent, and when it was first held.
- FR-16.2: The system shall show gains in green and losses in red.
- FR-16.3: The system shall display the portfolio's sector allocation, and selecting a sector shall filter the positions list to that sector.
- FR-16.4: The system shall summarise, for each sector held, the value-weighted percentage move of those holdings today, labelled as illustrative market context based on the holdings' own price moves.

FR-17: Sector Simulation

- FR-17.1: The system shall simulate adding 5% of the portfolio's value to a chosen sector, in proportion to that sector's existing holdings, and shall show the health score before and after and the change in each subscore.
- FR-17.2: When the largest sector is at or above the high-concentration threshold, the system shall simulate moving the excess down to the low threshold into the smallest sector and shall show the health change and the before and after allocation.
- FR-17.3: The system shall state that each simulation is analysis only and not a trade instruction, and shall not change any portfolio data.
- FR-17.4: The system shall offer simulations only when the portfolio spans at least two sectors.

4.4 Subsystem 4: Unusual Moves and News

FR-18: Unusual-Move Detection

- FR-18.1: When the dashboard loads, the system shall scan the daily closing prices of each holding in the most recently imported portfolio for unusual moves, and shall keep the result for 15 minutes.
- FR-18.2: The system shall compute daily log returns, ignoring returns across gaps of more than four calendar days, and shall skip any holding with fewer than 60 usable returns.
- FR-18.3: The system shall estimate each holding's volatility with an EWMA (λ 0.94), seeded from the variance of its first 30 returns and using only returns before the day being scored, and shall flag a day when the absolute return is at least three times that volatility.
- FR-18.4: The system shall grade flagged days as unusual (3 or more times), very unusual (4 or more times) or extremely unusual (5 or more times).
- FR-18.5: The system shall mark up to eight flagged days, largest first, on the portfolio line of the performance chart, red for falls and green for rises.
- FR-18.6: The nightly job shall record every detected move in a price-anomaly register and mark it invalid when it is caused by a price gap, a cents-to-rand unit flip (a price ratio within 5% of 100 or 1/100) or a spike of 20% or more that reverses by at least 80% the next day.

FR-19: Unusual-Move Explanation

- FR-19.1: When the user selects a marker, the system shall show the holding, the move, the date and the band and, for a holding in the built-in instrument catalogue (FR-8.1), the move's impact on the portfolio: the holding's weight on the day before (or today's weight when that day's value is unknown) multiplied by the move.
- FR-19.2: The system shall state the band, the move as a multiple of the holding's normal daily move and, when it is the largest in the period, that fact.
- FR-19.3: For a holding in the built-in instrument catalogue (FR-8.1), the system shall run a market-model event study against the holding's regional benchmark: beta and alpha estimated by ordinary least squares over trading days 120 to 21 before the event (at least 60 observations), abnormal return = actual return - beta x benchmark return, and the cumulative abnormal return from 5 days before to 10 days after the event with a 95% band of $1.96 \times \sigma \times \text{square root of the number of days}$.
- FR-19.4: For such a holding, the system shall split the move into a market part and a company-specific part and describe it as mainly company-specific, mainly market-driven, mixed or against the market.
- FR-19.5: For such a holding, when at least ten other tracked tickers were scanned for that date, the system shall state how many also moved unusually in the same direction if at least three did (naming up to three), and shall say so when none moved unusually and at least twenty were scanned.
- FR-19.6: For such a holding, the system shall list up to five stored articles tagged with the holding, published between three days before and one day after the move in the exchange's local dates, that have a provider match score and either a score of at least 25 or a headline containing the ticker or any word of the holding's name. They shall be ranked by $0.5 \times \text{BM25 relevance (divided by the query terms' total IDF, capped at 1)} +$

$0.3 \times \exp(-d^2 / 8)$, where d is the days between publication and the move, + 0.2 x (1 if the headline names the holding, otherwise 0.5).

- FR-19.7: Each listed article shall show its headline (opening the source in a new tab), a Closely related or Related label, source, date, how many days before or after the move it was published and the provider's highlighted sentence, followed by a statement that publication near the move does not mean the article caused it.
- FR-19.8: When no article qualifies, the system shall show "No matching news found" with an explanation.
- FR-19.9: The system shall offer the detection and event-study details (detector settings, beta, alpha, R squared, windows, the cumulative abnormal return chart with its band, the working and per-article scores) and a button that asks the AI assistant to explore possible causes.
- FR-19.10: For a holding outside the built-in instrument catalogue, the system shall not show the impact, run the event study, split the move, count same-day moves or search for articles, and shall show "No matching news found".

FR-20: News Collection

- FR-20.1: The system shall store news articles from the news provider and link an article to a ticker only when the provider tags it with that ticker as an equity listed in the ticker's country.
- FR-20.2: When the News page requests a held ticker's news, the system shall refresh it from the provider only if there has been no successful refresh for that ticker in the last six hours, requesting up to three articles, and shall pause these requests for 15 minutes after three consecutive failures.
- FR-20.3: The system shall run a nightly job at 00:30 UTC (02:30 SAST) that refreshes prices, records unusual moves, fetches the latest news published since the start of the last nightly run that ended ok or partial (or the last 36 hours), then fetches news published from four days before to two days after each recent validated unusual move that has no matching stored article and has not already been fetched, and then, budget permitting, catches up on older validated moves from the last 183 days.
- FR-20.4: The nightly job shall make at most 60 news requests per run and shall keep total news requests within 80 per UTC day.
- FR-20.5: The nightly job shall allow only one run at a time and shall record each run's status, counts and errors.

FR-21: News Page

- FR-21.1: The system shall display, on the My portfolio tab, up to 20 stored articles for each ticker held across the user's portfolios, newest first, each with its image, headline, description, source, date, ticker badge and a sentiment label (positive, negative or neutral) from the provider's sentiment score.
- FR-21.2: The system shall display counts of relevant, positive, negative and neutral articles.
- FR-21.3: The system shall allow the user to filter articles by ticker.
- FR-21.4: The system shall display general market news on the All Market tab, by category (Business, Top, Technology, Politics or Crime).
- FR-21.5: The system shall open each article's source in a new browser tab from its Read article link.

- FR-21.6: The system shall show "No holdings found" with an Import Holdings button when the user has no holdings, and "No news available" when no articles exist.

4.5 Subsystem 5: Analytics and Financial Indicators

FR-22: Indicator Calculation

The system shall calculate the following seven indicators for each distinct ticker held across all of the user's portfolios, from cached daily closing prices and company fundamentals.

- FR-22.1: The system shall calculate the CAPM expected return as risk-free rate + beta x (market return - risk-free rate), with a risk-free rate of 2% and a market return of 8%.
- FR-22.2: The system shall calculate the P/E ratio as the latest closing price divided by trailing earnings per share when earnings are positive, and otherwise use the provider's trailing or forward P/E; for a fund with no P/E available, it shall show N/A with the reason that funds do not report company-level financials.
- FR-22.3: The system shall calculate the Altman Z-score ($1.2 \times \text{working capital}/\text{total assets} + 1.4 \times \text{retained earnings}/\text{total assets} + 3.3 \times \text{EBIT}/\text{total assets} + 0.6 \times \text{market value of equity}/\text{total liabilities} + 0.999 \times \text{sales}/\text{total assets}$) and classify it as safe (above 2.99), grey zone (1.81 to 2.99) or distress (below 1.81), showing N/A with a reason for funds and whenever the required financial statement items are unavailable (with a note when the company is a bank or other financial institution).
- FR-22.4: The system shall calculate Beta as the covariance of the holding's daily returns with those of a global equity market proxy (the VT exchange traded fund) divided by the variance of the proxy's returns, over their common trading days, requiring more than 10 of them.
- FR-22.5: The system shall calculate the 14-day RSI with Wilder smoothing and classify it as overbought (above 70), neutral (30 to 70) or oversold (below 30).
- FR-22.6: The system shall calculate the Sharpe ratio as $(\text{mean daily return} \times 252 - 4\%) / (\text{standard deviation of daily returns} \times \text{square root of } 252)$, requiring more than 10 daily returns.
- FR-22.7: The system shall calculate the Sortino ratio as the Sharpe ratio but using the standard deviation of negative daily returns only.

FR-23: Indicator Presentation

- FR-23.1: The system shall display one card per holding with one cell per selected indicator, showing the value to two decimal places and a short classification.
- FR-23.2: The system shall show a one-line plain-language explanation when the pointer rests on an indicator's name, and a dialog with why it matters, how it is worked out, its formula, any sector note and its value across all holdings when the user selects a value.
- FR-23.3: The system shall show N/A with a short reason in the cell of an indicator that cannot be calculated, without affecting the other indicators, and Error in every cell of a holding that has no price history.
- FR-23.4: The system shall colour a value green when it meets the indicator's favourable threshold and red when it meets its unfavourable threshold.
- FR-23.5: The system shall allow the user to choose which indicators are shown, per holding or for all holdings, using presets (Most Popular, Value Investing, Risk & Volatility, Technical / Momentum, All Indicators) or their own named presets, and shall remember the choice in the browser.

- FR-23.6: The system shall show "No Holdings" with an Import Holdings button when there are no holdings, and "Failed to load indicators" when the request fails.

4.6 Subsystem 6: Market Data

FR-24: Market Data Provision

- FR-24.1: The system shall store daily price bars (open, high, low, close, previous close and volume) per ticker and serve them from this store, refreshing a ticker from the provider only when its data was fetched more than 24 hours ago or its latest bar is more than four days old.
- FR-24.2: The system shall store company fundamentals per ticker and refresh them after seven days.
- FR-24.3: The system shall convert JSE prices, which the provider quotes in cents, to rand.
- FR-24.4: The system shall allow the user to search for a share by ticker or company name and show up to ten matching shares, ETFs or indices, keeping results for five minutes.
- FR-24.5: When a provider is unavailable, the system shall continue with the most recent stored prices for holdings, indicators, benchmarks and unusual-move detection. Watchlist quotes are fetched live and are not stored.

4.7 Subsystem 7: AI Assistant

FR-25: Conversational Interface

- FR-25.1: The system shall provide a chat interface on the AI Assistant page and an assistant dock on every other signed-in page, where the user can ask natural-language questions.
- FR-25.2: The system shall combine each question with the user's remembered facts, a summary of earlier turns, and the portfolio context (the 15 largest holdings of each portfolio with sector, quantity, average cost, value and gain, and the health score and subscores) before it reaches the AI model.
- FR-25.3: The system shall let the model call tools that fetch share prices, analytics indicators, market news, statement detail, ticker matches and a goal projection (a Monte Carlo simulation of 2,000 paths), with up to four tool rounds per question.
- FR-25.4: The system shall allow the user to limit the conversation to one portfolio or to all portfolios.
- FR-25.5: The system shall stream each new answer as formatted text and, on the AI Assistant page, allow the user to copy an answer and regenerate the last one; a regenerated or retried answer is returned whole rather than streamed.
- FR-25.6: The system shall display three suggested prompts on the AI Assistant page when the conversation is empty, which the user can send with a single click.
- FR-25.7: The system shall provide ready-made questions from dashboard cards (EquityLens Insight, which opens a menu of questions), from insights (Ask AI Why, inside the Why? reveal) and from event cards (Explore possible causes); choosing one opens the dock and sends the question.
- FR-25.8: The system shall show "Something went wrong, try again." when an answer cannot be generated, with a Retry button on the AI Assistant page.

- FR-25.9: The system shall limit each user to 5 messages in 60 seconds and 50 messages per UTC day, and shall show the time until the next message is allowed.

FR-26: Conversation Management

- FR-26.1: The system shall save the user's conversations so that they persist across sessions.
- FR-26.2: The system shall list the user's saved conversations, most recently active first, and load a conversation's full message history when selected.
- FR-26.3: The system shall allow the user to start a new conversation at any time.
- FR-26.4: The system shall give each new conversation an automatic title of at most five words.
- FR-26.5: The system shall allow the user to rename a saved conversation.
- FR-26.6: The system shall allow the user to delete a saved conversation.

FR-27: AI Memory

- FR-27.1: After each answer, the system shall extract up to five lasting facts about the user's goals, time horizon, risk appetite, constraints and preferences from the user's message, keeping at most 30 facts of at most 300 characters each, and shall instruct the extraction model not to keep sensitive personal information (health, family, identity numbers, contact details).
- FR-27.2: The system shall include the stored facts in every conversation.
- FR-27.3: The system shall allow the user to view the stored facts and delete any of them.

FR-28: Guidance and Disclaimers

- FR-28.1: The system shall instruct the AI model to provide education and analysis rather than tell users to buy or sell specific securities, and to explain the trade-offs instead.
- FR-28.2: The system shall instruct the AI model to help only with the user's portfolio and statements, using EquityLens, investing education, share performance and risk, market news and financial goals, to decline other requests (including code, homework and role-play), and to treat portfolio data and tool results as data rather than instructions.
- FR-28.3: The system shall display a persistent disclaimer that AI answers are informational only and not financial advice, on the AI Assistant page and in the assistant dock.

4.8 Subsystem 8: Watchlist

FR-29: Watchlist Management

- FR-29.1: The system shall allow the user to add a share they do not own to their watchlist by searching for it by ticker or company name.
- FR-29.2: The system shall reject an instrument whose provider quote type is not a stock, ETF or index, showing the reason.
- FR-29.3: The system shall display the first three watchlist entries, each with its ticker, company name (or the ticker when no name is stored), the latest price prefixed "R" (JSE prices converted from cents to rand; other listings shown in their own currency) and daily percentage change.

- FR-29.4: The system shall allow the user to remove an entry from their watchlist.
- FR-29.5: The system shall keep each user's watchlist private to that user.

4.9 Subsystem 9: Explore

FR-30: Similar-Share Suggestions

- FR-30.1: The system shall suggest up to nine JSE shares that are similar to the JSE-listed shares the user holds across their portfolios, excluding shares already held.
- FR-30.2: The system shall draw suggestions from a universe of up to 120 of the largest JSE shares with a market capitalisation of at least R1 billion, rebuilt at most once every seven days.
- FR-30.3: The system shall measure similarity as the distance between standardised values of market capitalisation (log scale), free float and dividend yield (capped at 15%), adding 0.4 when the sectors differ, and shall report closeness as a percentage match.
- FR-30.4: The system shall explain each suggestion, naming the held share it matches, whether the sector is the same and the traits that are similar.
- FR-30.5: The system shall plot holdings and suggestions on a map of market capitalisation against free float, sized by dividend yield and coloured by sector.
- FR-30.6: The system shall list holdings that are not JSE shares as not included.

4.10 Subsystem 10: General Interface

FR-31: General Interface

- FR-31.1: The system shall provide a sidebar with Dashboard, Portfolio, Analytics, News, AI Assistant, Explore, Settings and Help.
- FR-31.2: The system shall allow the user to switch between light and dark themes on the Settings page and remember the choice in the browser.
- FR-31.3: The system shall allow the user to blur all rand values from the top bar and remember the choice in the browser.
- FR-31.4: The system shall provide a Help Centre with quick links to the Dashboard, Portfolio, Analytics, News and AI Assistant, a usage tip, three frequently asked questions, the Excel template and a support e-mail link.
- FR-31.5: The system shall provide a public landing page that describes the product and links to registration and login.
- FR-31.6: The system shall provide a skip-to-content link and a page for unknown addresses.

5. Non-Functional Requirements

5.1 NFR-1: Security

- NFR-1.1: 100% of API endpoints, other than the health check and the registration/login/verification endpoints, shall reject unauthenticated requests with an HTTP 401 response.
- NFR-1.2: User credentials shall never be stored in plain text by the system. Sessions shall be established with expiring access tokens with a lifetime of at most 60 minutes; an expired session shall require re-authentication or a silent token refresh before any protected resource is served.
- NFR-1.3: Multi-factor authentication (TOTP) shall be enforced on user accounts.
- NFR-1.4: All data in transit shall be encrypted with TLS;
- NFR-1.5: All portfolio data shall be scoped to the owning user at the data-access level: 0 instances of cross-user data exposure.
- NFR-1.6: All portfolio data shall be scoped to the owning user at the data-access level: 0 instances of cross-user data exposure, verified by integration tests.
- NFR-1.7: All personal data handling shall comply with POPIA.

5.2 NFR-2: Performance

- NFR-2.1: API endpoints shall respond within 500 ms at the 85th percentile under normal load.
- NFR-2.2: Initial page load shall complete within 10 seconds on a standard broadband connection.
- NFR-2.3: Dashboard visualisations shall render within 6 seconds
- NFR-2.4: Market data shall be served through a cache-aside store: a request for an instrument that is already cached shall be answered with zero calls to the upstream provider, keeping repeat dashboard loads within provider rate limits.

5.3 NFR-3: Reliability

- NFR-3.1: The failure of any single external data source (market data, news, AI model) shall degrade only its own feature; the remainder of the system shall stay fully functional.
- NFR-3.2: Show a valid error message if the AI feature is not working
- NFR-3.3: A failed fetch for one holding's market data shall not prevent the dashboard from rendering the remaining holdings.

5.4 NFR-4: Scalability

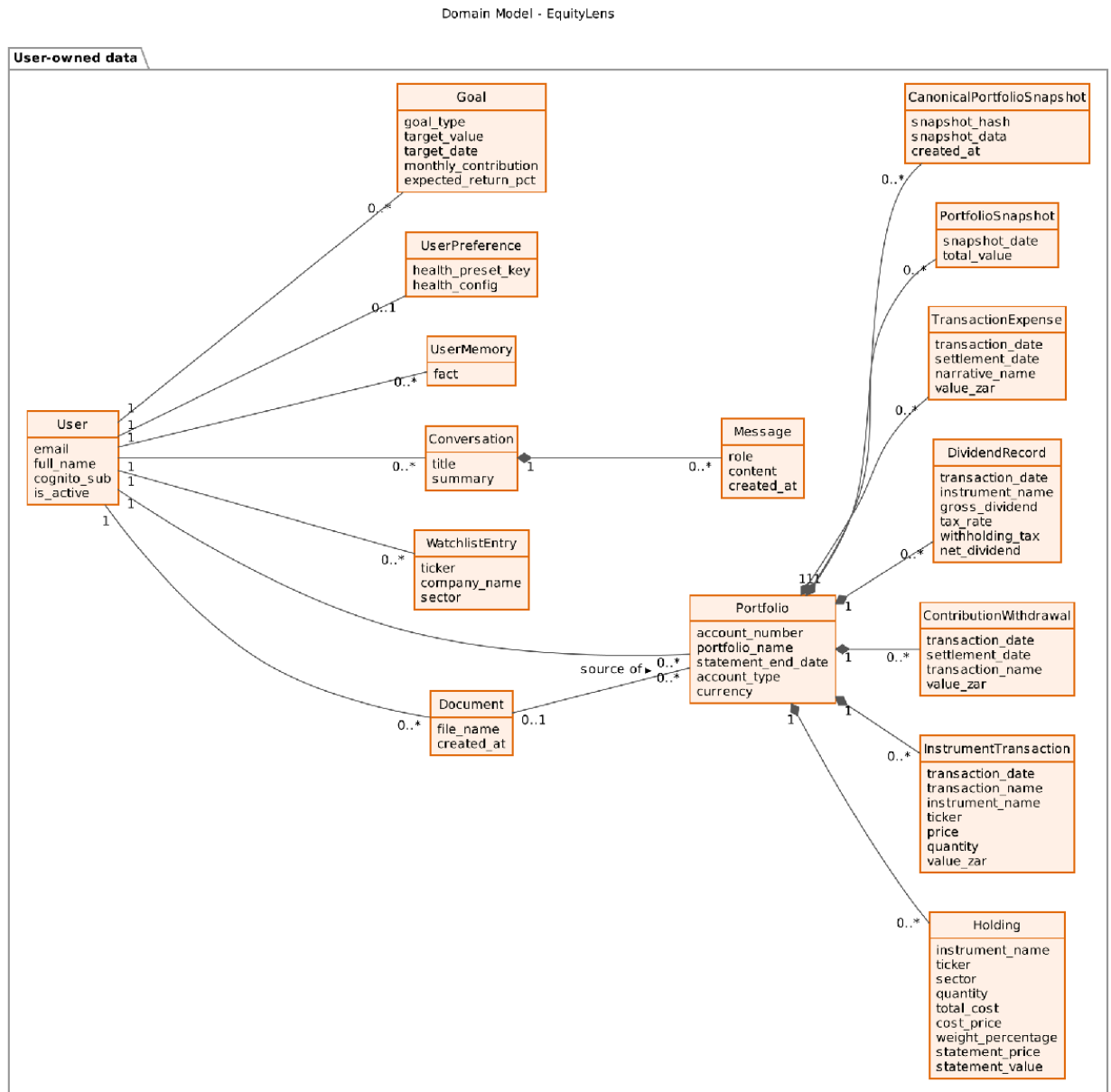
- NFR-4.1: The backend shall support at least 50 concurrent users without performance degradation beyond the NFR-2 targets.
- NFR-4.2: Send 100 simultaneous API Requests to EquityLens
- NFR-4.3: The system shall accommodate a 200% workload increase without major architectural changes and with no more than a 10% performance decrease.

5.5 NFR-5: Usability and Accessibility

- NFR-5.1: A new user shall be able to register, import a portfolio and reach a populated dashboard within 5 minutes of first use.
- NFR-5.2: The interface shall meet WCAG 2.1 AA: all interactive elements keyboard-navigable with visible focus states, and text contrast ratios of at least 4.5:1.
- NFR-5.3: Every financial indicator shall have a plain-language explanation accessible from where the indicator is displayed, so no finance training is required to use the system.
- NFR-5.4: The interface shall be fully responsive across desktop (1280px+), tablet (768-1279px) and mobile (320-767px) with no horizontal scrolling on mobile.

6. Domain Model

6.1 UML Class Diagram



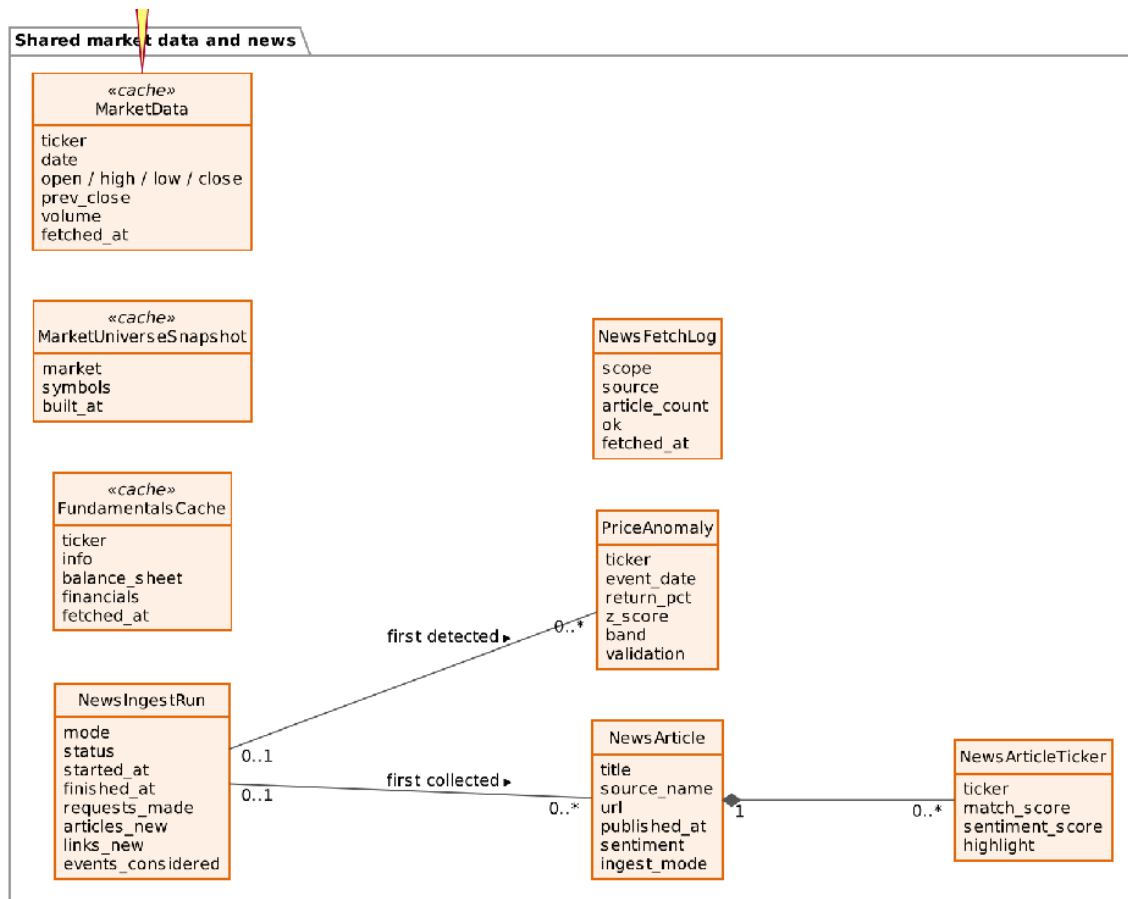


Figure 6.1 - Domain model

6.2 Entity Descriptions

User - an account holder. `cognito_sub` links the local record to the identity provider; no password is stored for identity-provider accounts. Every other user-owned record (documents, portfolios, watchlist entries, conversations, AI memories, preferences and goals) references the user and is read only through that user's account.

Document - a record of an imported statement file: its file name and when it was imported. The file's content is not stored.

Portfolio - one imported statement: account number, portfolio name, statement end date, account type (zar, tfsa or usd) and currency. A user may hold many portfolios; the dashboard uses the most recently imported one.

Holding - a position within a portfolio: instrument name, resolved ticker and sector, quantity, total cost, per-share cost price, weight, and the statement-date price and value. Current value is derived at read time from market data, not stored.

InstrumentTransaction - a purchase or sale recorded in the statement, with date, type, instrument, ticker, sector, price, quantity and rand value.

ContributionWithdrawal - cash paid into or withdrawn from the account, with transaction and settlement dates.

DividendRecord - a dividend received per instrument: gross amount, tax rate, withholding tax and net amount.

TransactionExpense - a fee or cost recorded in the statement.

PortfolioSnapshot - at most one row per portfolio per day holding the portfolio's total value, written for the portfolio the dashboard shows (days rebuilt from its transactions, plus each day the dashboard is opened) and used to build the performance history. Uniqueness is enforced on (portfolio, date). The table has a benchmark_value column, but it is always left empty; the benchmark is recomputed from price history.

CanonicalPortfolioSnapshot - a stored snapshot of a portfolio's summary, analytics and news, identified by its SHA-256 hash, from which the PDF report is generated.

WatchlistEntry - a share the user tracks without owning it: ticker, company name and sector. Private to the user.

Conversation - a saved AI assistant conversation with a user-editable title and a running summary of older turns.

Message - a single message within a conversation; role distinguishes the user's messages from the assistant's.

UserMemory - a fact about the user's goals or preferences that the assistant keeps across conversations (at most 30 per user).

UserPreference - the user's health-score yardstick: a preset key or custom settings.

Goal - a financial goal (type, target value and date, monthly contribution, expected return). The table exists and is read when suggesting a yardstick preset, but the current release has no screen that creates goals.

MarketData (cache) - a daily price bar per ticker and date (open, high, low, close, previous close and volume) with its fetch time, retained to serve repeat requests. It is a cache, not a system of record.

FundamentalsCache - company information, balance sheet and income statement per ticker, with its fetch time.

NewsArticle - a stored news article: provider id, title, description, URL, image, source, publication time, sentiment and how it was collected.

NewsArticleTicker - the link between an article and a ticker the provider tagged, with the provider's match score, sentiment and highlighted sentence.

NewsFetchLog - one row per MarketAux request made through the news-ingest service (News page refreshes and the nightly and backfill jobs), used for the six-hour refresh rule and the daily request budget. The AI assistant's news tool, the PDF report and NewsData.io requests are not logged.

NewsIngestRun - one row per nightly or backfill news run, with its status and counts.

PriceAnomaly - a detected unusual move per ticker, date and detection threshold: return, z-score, volatility, direction, band and validation result.

MarketUniverseSnapshot - the cached list of JSE shares used by Explore, with the time it was built.

7. Requirements-Use Case Traceability

The matrix marks, for each functional requirement group, the use cases that it helps realise. Every use case is realised by at least one requirement group, and every requirement group is exercised by at least one use case.

FR group	UC-1	UC-2	UC-3	UC-4	UC-5	UC-6	UC-7	UC-8	UC-9	UC-10	UC-11	UC-12	UC-13	UC-14	UC-15	UC-16	UC-17	UC-18	UC-19	UC-20	UC-21	UC-22	UC-23	UC-24	UC-25	UC-26
FR-1 User Registration	X																									
FR-2 Email Verification		X																								
FR-3 Login and Session			X		X					X																
FR-4 Multi-Factor Authentication			X																							
FR-5 Password Reset				X																						
FR-6 Account Deletion						X																				
FR-7 Statement Import							X																		X	
FR-8 Instrument Resolution							X																			
FR-9 Portfolio Summary								X																		
FR-10 Portfolio Snapshot Report								X	X																	
FR-11 Data Ownership						X	X	X		X																
FR-12 Dashboard Overview										X																
FR-13 Portfolio Insights													X													
FR-14 Portfolio Health													X	X	X											
FR-15 Performance vs Benchmark											X															
FR-16 Positions and Sectors										X																
FR-17 Sector Simulation															X											
FR-18 Unusual-Move Detection												X														X
FR-19 Unusual-Move Explanation												X														
FR-20 News Collection												X						X								X
FR-21 News Page																		X								
FR-22 Indicator Calculation																	X									
FR-23 Indicator Presentation																	X									
FR-24 Market Data Provision										X	X					X	X						X			X
FR-25 Conversational Interface																		X	X							
FR-26 Conversation Management																		X		X						
FR-27 AI Memory																		X			X					
FR-28 Guidance and Disclaimers																		X	X							
FR-29 Watchlist Management															X											
FR-30 Similar-Share Suggestions																							X			
FR-31 General Interface																								X	X	

8. Assumptions and Dependencies

8.1 Assumptions

- A-1: Users have basic familiarity with share tickers but no formal finance training.
- A-2: Users hold, or intend to build, an investment portfolio with at least one holding.
- A-3: Users have a modern web browser with JavaScript enabled, and an authenticator app on a phone or computer for TOTP.
- A-4: Imported PDF statements are EasyEquities statements. Data from any other source is entered in the EquityLens Excel template, with values in rand and dates as yyyy-mm-dd.
- A-5: Holdings are valued with end-of-day data (the latest stored daily closing price), while watchlist quotes use the provider's current price; the provider quotes JSE prices in cents.
- A-6: A holding is assigned a benchmark region only when it appears in the built-in instrument catalogue; other holdings are left out of the benchmark weighting.
- A-7: Unusual moves are statistical flags, and articles published near a move are shown as context, not as proven causes.

8.2 Dependencies

- D-1: Identity provider (Amazon Cognito) for registration, email verification, sign-in, TOTP and password reset. If unavailable: new sign-ins and registrations fail; signed-in users keep working while their access tokens are valid, because the server verifies tokens locally with the Cognito signing keys it has already downloaded; if the backend restarts during the outage, it cannot download the keys and authenticated requests fail.
- D-2: Market data providers (Yahoo Finance, and Alpha Vantage for non-JSE tickers when a key is configured) for prices, fundamentals, search, exchange rates and the Explore universe. Yahoo price refreshes and company fundamentals are used only when the ALLOW_LIVE_MARKET_FALLBACK setting is on (benchmark, exchange-rate, AI price-tool, PDF report and nightly price fetches always use the live path); with it off, JSE prices are refreshed only by those always-live fetches and fundamentals-based results (P/E, Altman Z-score, Explore) are empty. If a provider is unavailable: the last stored prices are used; holdings without a price are valued at statement price, then cost; search, watchlist quotes and new Explore universes are unavailable.
- D-3: News providers (MarketAux for holdings news, the nightly job, the AI assistant's news tool and the PDF report; NewsData.io for general market news). If unavailable: previously stored articles are still shown and unusual-move explanations still use stored news; new articles are not collected; nothing else is affected.
- D-4: AI model provider (Amazon Bedrock, Anthropic Claude models) for answers, conversation summaries, titles and memory extraction. If unavailable: the AI Assistant page shows a generic error with a Retry button (the assistant dock shows the error without Retry); other features are unaffected.
- D-5: AWS hosting in the Africa (Cape Town) region: a single EC2 instance (Terraform defaults to the free-tier-eligible t3.micro; the production size is not recorded in the repository), an RDS PostgreSQL database, ECR, Secrets Manager and Systems Manager, with Amazon Bedrock billed per use. Secrets Manager and Bedrock are

charged outside the AWS Free Tier, so spending is kept within the project's budget of about \$100 over two months. If spending exceeds that budget: the project's cost constraint is violated.

- D-6: GitHub Actions for CI quality gates, deployment and the nightly news schedule. If unavailable: automated checks and deploys stop, and the nightly news collection does not run (holdings news is still refreshed when the News page is opened).
- D-7: In-browser file parsing (pdf.js for PDF statements, SheetJS for Excel). If pdf.js cannot load: PDF import is unavailable; Excel import still works.
- D-8: Let's Encrypt certificates for HTTPS. If a certificate lapses: browsers reject the site until it is renewed.